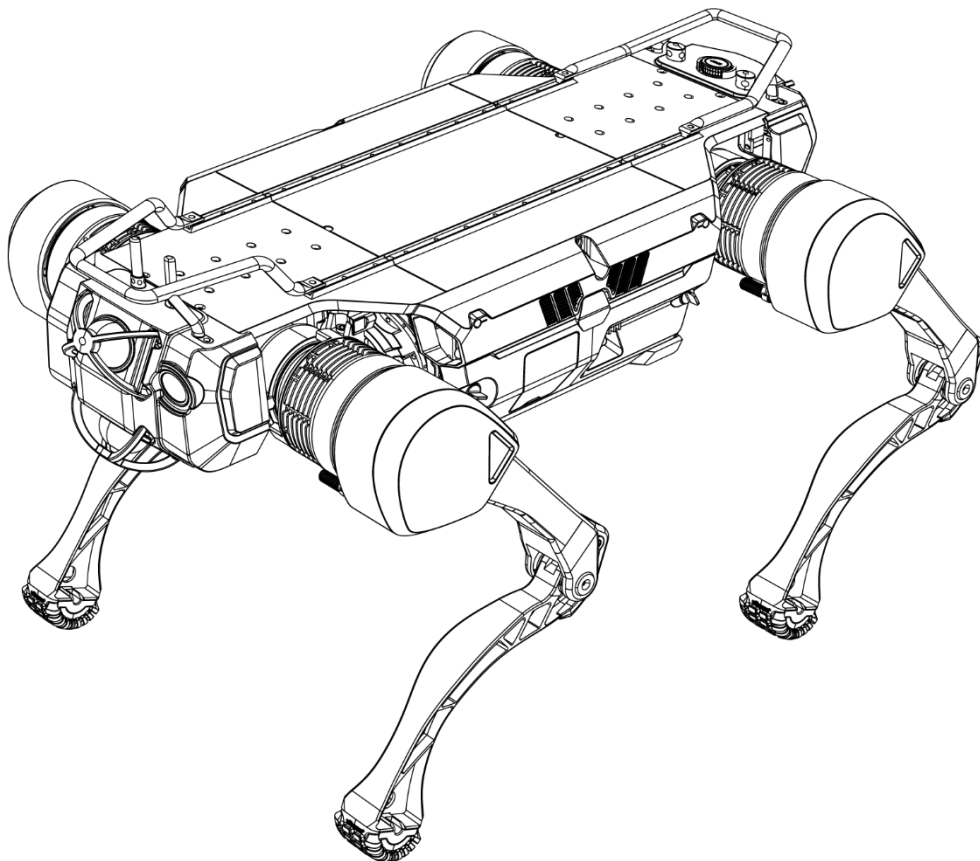


Jueying X30 Pro Application Manual

V2.0.4-0 2025.11.29



Revision History

Version	Date	Note
0.0.4-0	2023/10/25	First version
0.0.5-0	2024/2/23	Delete command timeout, modify lidar angle and unit
0.0.8-0	2024/6/14	Add multi-floor map and navigation, waypoint record, LiDAR FOV; modify lidar data rostopic
2.0.0-0	2024/8/16	Update the mapping procedure for slam package version 2.1.0 and later.
2.0.1-0	2025/4/7	Update the coordinate of sensors (≥ 401)
2.0.3-0	2025/5/8	Update usage of Navigation Recorder
2.0.4-0	2025/11/29	Update the wiring definition for the USB 2.0

※The final interpretation right belongs to DEEP Robotics.

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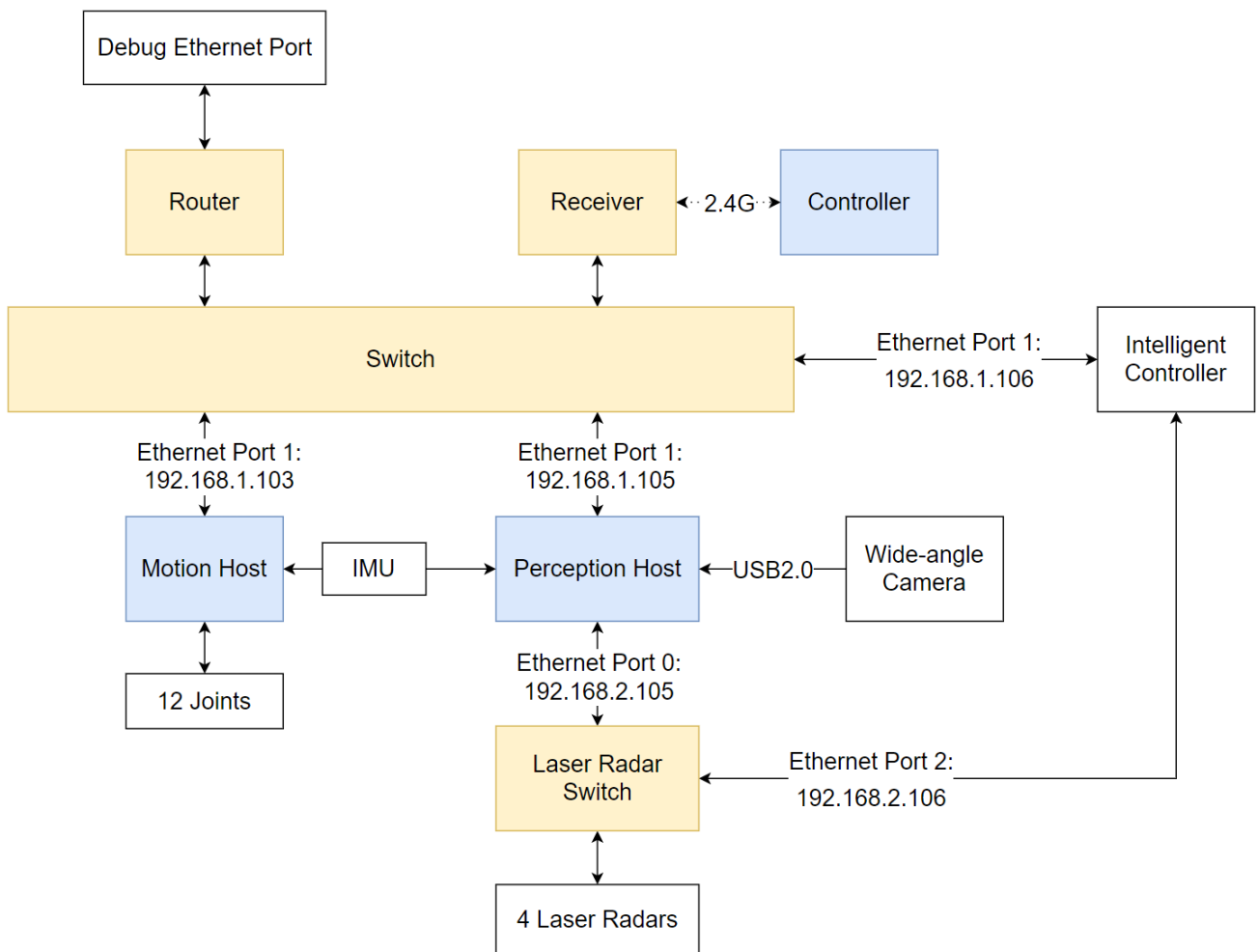
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1 Overall Architecture

Jueying X30 Pro is equipped with the motion host and the perception host internally, and the intelligent controller on its back has the navigation host built into it. The development host can be connected to each host of the robot through a wireless router. Each host has been configured with the relevant commonly-used software and drivers.



2 Motion Host

Jueying X30 Pro motion program opens the configuration permissions of some parameters, and provides the logging function.

2.1 Remote Login

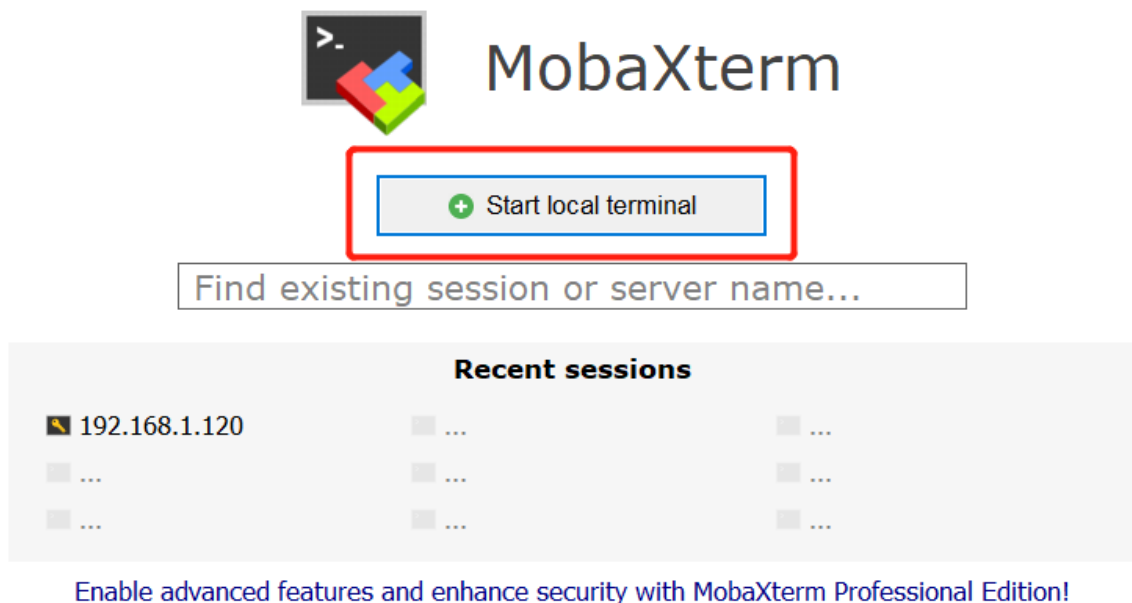
You can remotely log in to the motion host through the MobaXterm software.

1. Connect the development host to the Debug Ethernet Port at the tail of robot by network cable, or connect the robot through Wi-Fi:

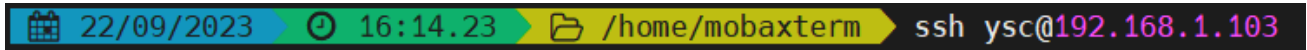
Wi-Fi name: X30***** (serial number of your robot)

Wi-Fi password is: 12345678

2. Open MobaXterm, and click the [Start local terminal] button.

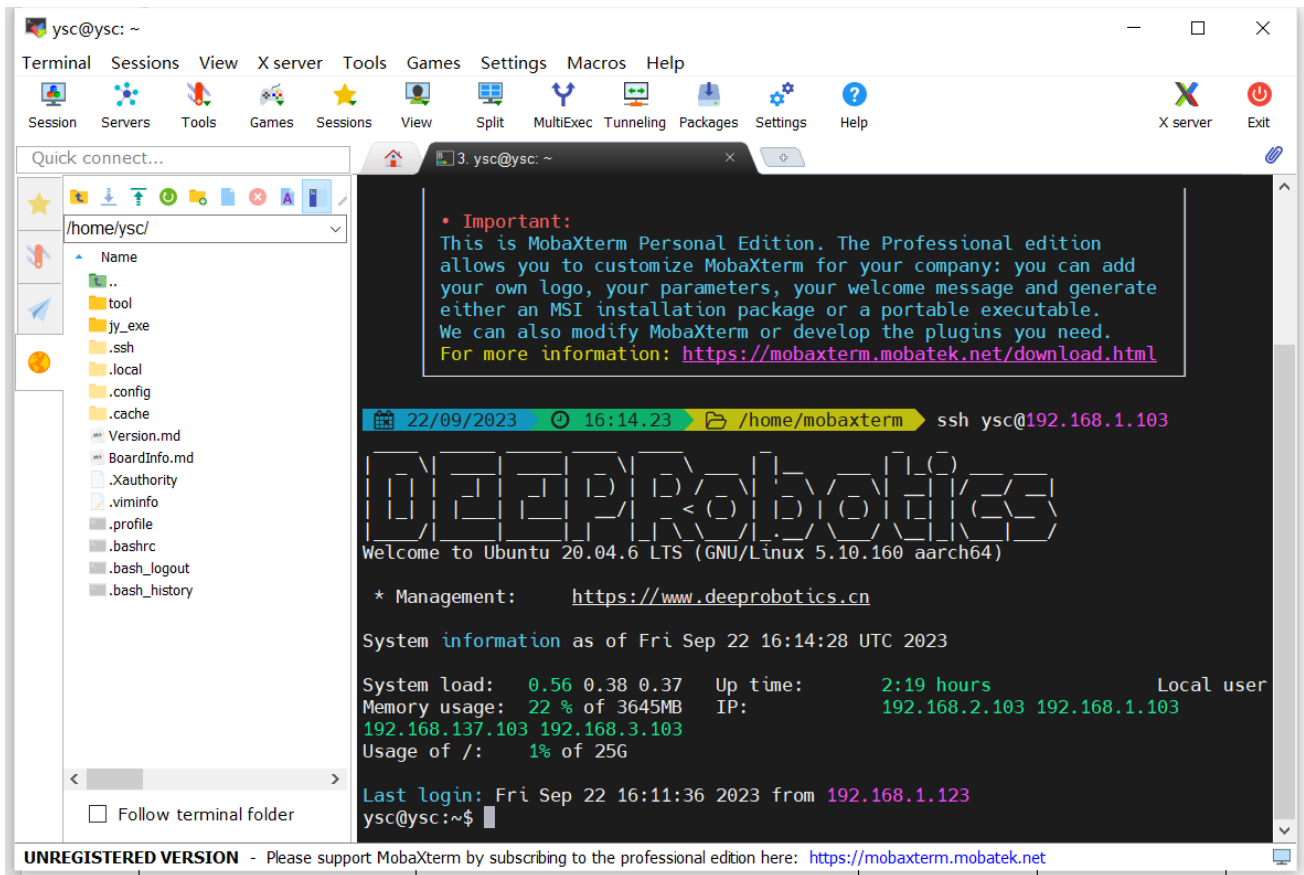


3. Enter `ssh ysc@192.168.1.103`, and the password is ' (a single quote).



2.2 Parameter Configuration

You can open the `/home/ysc/jy_exe/conf` folder in the left column of MobaXterm to modify the profile, and the modified configuration parameters will take effect after the restart of motion program.



motion.toml

Parameter	Meaning	Type	Range
load	Load weight (kg)	float	0.0~20.0
enable_twist	To enable user to control the robot to twist	bool	true/false
load_x	Coordinate value of load centroid on the X-axis (m)	float	-0.3~0.3

load_y	Coordinate value of load centroid on the Y-axis (m)	float	-0.1~0.1
load_z	Coordinate value of load centroid on the Z-axis (m)	float	0.0~0.3

overtime.toml			
[Wifi]: WiFi Connection Timeout			
Parameter	Meaning	Type	Range
StopMotion	Time from disconnection to stopping moving (sec)	int	>0
KeepStill	Time from disconnection to stopping stepping (sec)	int	>StopMotion
Sitdown	Time from disconnection to sitting down (sec)	int	> KeepStill

[Notes] When the robot is loaded, please do not turn on the enable_twist, otherwise it may cause the robot to fall down and damage the load.

[Notes] The modified parameter value must comply with the specified data type.

[Notes] If it needs to add or remove the load, please refer to "4.3 Motion Parameter Configuration" to modify configuration.

2.3 Logging Function

Jueying X30 Pro generates three types of log files during motion:

Event log (.log): Record each event during the running process. The file is stored in /home/ysc/jy_exe/log directory and can be opened from the file directory in the left column of MobaXterm.

Operation log (.csv): Press the [Save Data] button on the Settings page of the controller APP to save the motion state data recorded within the last 5 minutes, with the recording frequency of 200Hz. The data saved every time can be viewed directly in Microsoft Excel.

The file is stored in the same folder as the fault snapshot. The specific path is `/home/ysc/jy_exe/data/`, which can also be obtained at the end of the event log.

Fault snapshot (.snapshot.csv): Press the [Save Data] button on the Settings page of the APP to save the fault snapshot (that is, the data within the last 5 minutes), which takes 9 seconds to save. The file is stored in the same folder as the operation log. The specific path is `/home/ysc/jy_exe/data/`, which can also be obtained at the end of the event log.

[Notes] The event log is stored for 7 days (that is, 168 hours). Every time the program is run, the expired logs are cleaned up to prevent the hard disk space is not sufficient.

[Notes] When pressing the [Save Data] button, please ensure that the robot is lying down.

2.4 Motion Program Script

There are some scripts built into the `/home/ysc/jy_exe/scripts` path of Jueying X30 Pro motion host:

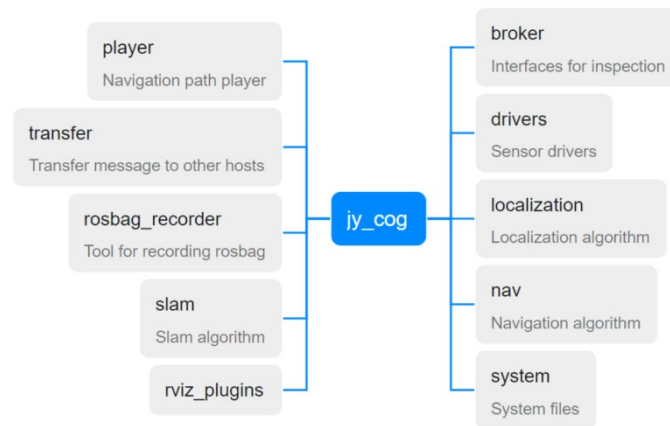
Script Name	Function
restart.sh	Restart the motion program
stop.sh	Quit the motion program
show_log.sh	View the real-time event log of motion program

You can enter the following command line in the terminal to run the script:

```
1 | cd ~/jy_exe/scripts
2 | ./xxxxx.sh # xxxxx.sh is the name of the script you want to run
```

3 Navigation Host

Jueying X30 Pro intelligent controller has the navigation host built into it, which is responsible for mapping, localization, and navigation. The directory `/home/ysc/jy_cog` in navigation host contains the following algorithms, drivers, applications, logs, startup scripts, and profiles related to the perceptive functions, with a directory structure as follows(in newer versions, *broker* has been renamed *robot_server*):



Among them, the `/jy_cog/system` directory contains the folders named `conf`, `scripts`, and `map`. The folder `conf` stores the profiles which can be used to configure the system modes, function parameters, etc; the `scripts` folder stores the script file; the `map` folder stores the maps.

3.1 Remote Login

You can remotely log in to the navigation host through VNC.

3.1.1 Use VNC on Windows PC

1. Download and install [MobaXterm\(Installer edition\)](#) on your own Windows PC.

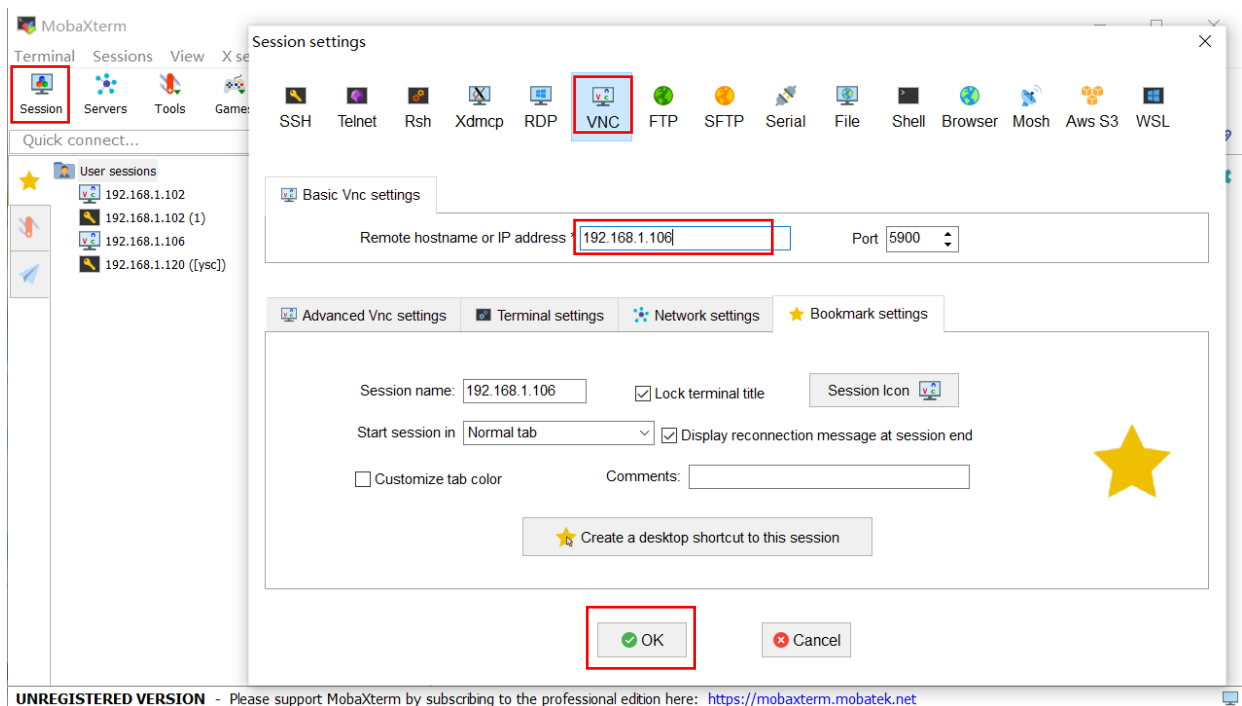
2. Connect the Windows PC to the Wi-Fi of robot:

WiFi Name: X30***** (serial number of your robot)

WiFi Password: 12345678

3. Open CMD tool, then enter `ping 192.168.1.106` to confirm that the navigation host is connected.

4. Open MobaXterm to configure VNC session:



① Click Session

② Select VNC

③ Enter IP: 192.168.1.106

④ Click OK

5. Click the newly created VNC session, enter the password, then remotely access the navigation host desktop.

IP: 192.168.1.106 Password: 123456

3.1.2 Use VNC on Ubuntu

1. Open a terminal on your Ubuntu host and enter the following command to install

Remmina:

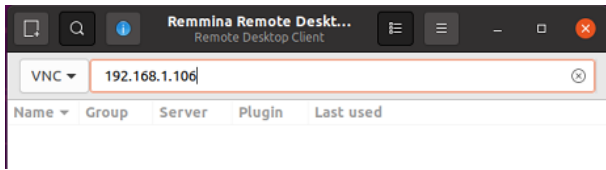
```
1 | sudo apt-get install remmina remmina-plugin-* -y
```

2. Connect the Ubuntu host to the Wi-Fi of robot:

WiFi Name: X30***** (serial number of your robot)

WiFi Password: 12345678

3. Enter `remmina` on the terminal to run Remmina.
4. Select VNC protocol, enter the IP of the navigation host and press Enter, then enter the password in the pop-up window and click OK.



5. Then the remote desktop can be seen.

3.2 Autorun Configuration

You can set the self-start of the localization and navigation program in the `autorun_slam` file in the `ji_cog/system/conf` directory. If the file content is set to false, then after the robot is turned on, the localization and navigation program will not automatically run, and you need to start it manually. This applies to the deployment of map; If the file content is set to true, then after the robot is turned on, the localization and navigation program will automatically run. After you provides the initial localization, the robot can automatically run according to the deployed map and navigation tasks. The autorun configuration takes effect after the navigation host restarts.

3.3 Map Deployment(for earlier than v2.1.0)

[Notes] Before starting mapping, please follow the remote connection procedure to connect to host 106 and enter: `dpkg -l slam` in the terminal to check the version of the slam package. The map deployment method described in this section applies to slam(<2.1.0). If the version is v2.1.0 or later, please refer to section 3.4 for map deployment.

3.3.1 Notice for Use

1. During mapping, please do not manipulate the robot for strenuous exercise and do not block the laser radar, otherwise the mapping may fail.
2. Before mapping, open the terminal and enter the `top` command to check the computing resources. If any other program uses too many computing resources, enter the following command line to close it: `sudo kill [PID]`.

```

ysc@ysc-perception:~$ top
top - 10:43:20 up 6 min, 1 user, load average: 1.26, 0.81, 0.37
Tasks: 283 total, 2 running, 281 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.3 us, 4.0 sy, 0.0 ni, 89.1 id, 0.1 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 15872.0 total, 13393.9 free, 1389.6 used, 1088.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 14008.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 2175 ysc        20   0 436292 26328 10404 R   27.9   0.2   1:41.68 yesense+
 2127 ysc        20   0 518480 29676 13056 S   19.3   0.2   1:16.53 rslidar+

```

3. Make sure the mapping program is not running. If it is running, please save the map and exit the program first according to 3.3.2.1. Make sure the navigation program is not running. If it is running, please enter the following command line in the terminal to kill it:

```
1 | bash ~/jy_cog/system/scripts/kill_nav.sh
```

4. Before starting the mapping program, it needs to check if there are created maps in the `/jy_cog/system/map` folder. After the mapping program starts, this folder will be emptied. If necessary, please move the created maps to another folder in advance and

move them back when it needs to use them again.

- Currently, the mapping for narrow stair, outdoor extremely open area, or ultra-large venue is not supported.

3.3.2 Single-Floor Mapping

3.3.2.1 Start the Mapping Program

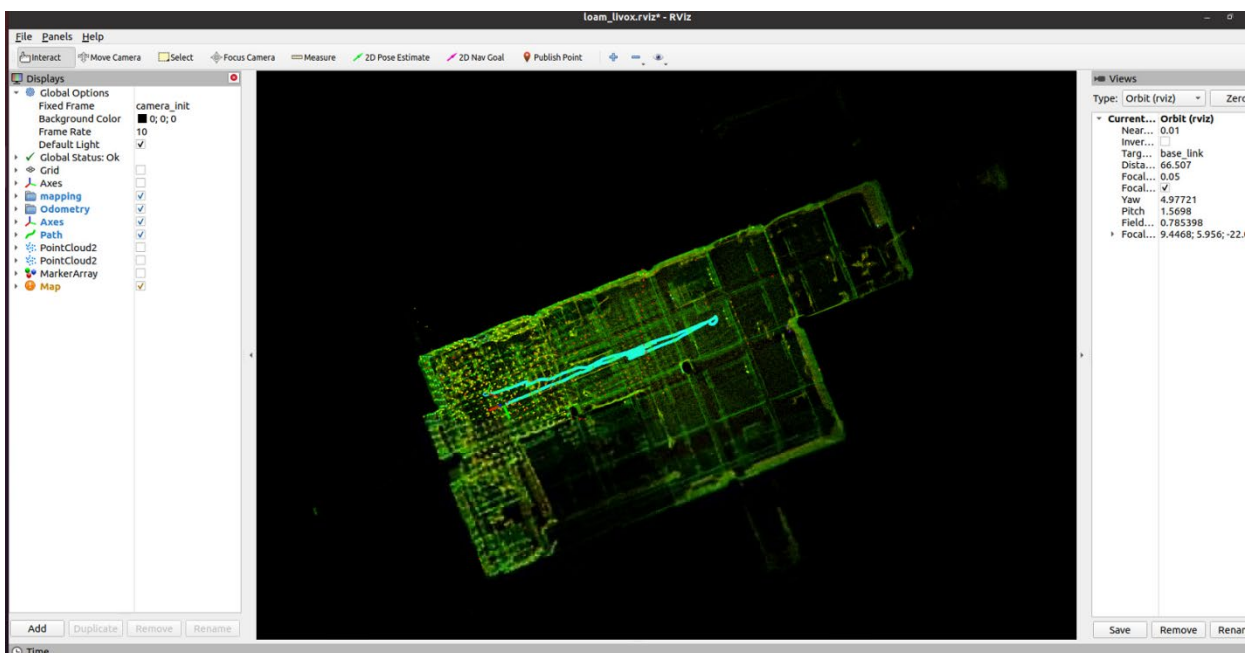
[Notes] This version of mapping program is suitable for the indoor scenes with an area not more than 200m×100m.

[Notes] Please start the mapping program when the robot is standing.

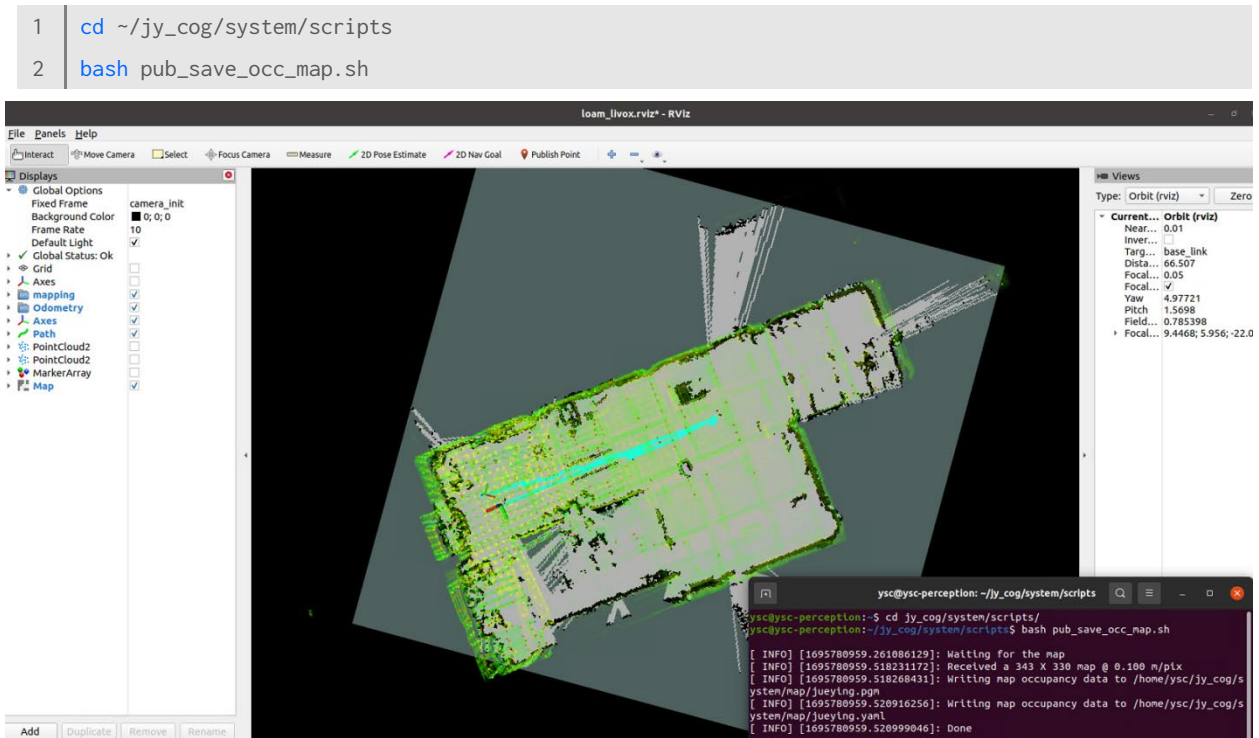
- Open the terminal, then enter the following commands to start the script of mapping:

```
1 cd ~/jy_cog/system/scripts
2 bash start_mapping.sh
```

- After starting the mapping program, please first keep the robot standing still for 5-8 seconds, then remotely control the robot to walk at constant speed smoothly and steadily in the area that needs to be mapped. During scanning, you can check whether the localization of robot is normal and whether the point cloud is drifting in RViz.



3. After the robot finishes the scanning of the mapping area, press **Ctrl+C** in the terminal which starts mapping to quit mapping. At the same time, the program automatically generates a grid-based map and save it to the `/jy_cog/system/map` folder. If it needs to preview the grid-based map during mapping, please enter the following commands in a new terminal:



[Notes] If the mapping program cannot be successfully closed, please run the following commands in the terminal to kill the mapping process:

```
1 | cd ~/jy_cog/system/scripts
2 | bash kill_mapping.sh
```

3.3.2.2 Offline Gridification

If the grid-based map cannot be obtained from the above steps, you can run the following commands after closing the mapping program to convert the point cloud map obtained by scanning to the grid-based map offline:

```
1 | cd ~/jy_cog/system/scripts
2 | ./trans_save_occ_map.sh
```




If the obtained grid-based map is not consistent with the actual environment, you can configure the parameters of grid-based map in the `params.yaml` file in the `jy_cog/system/conf/slam/pcd2map` folder.

Parameter	Meaning	Type
<code>minh</code>	The height of the lowest point cloud intercepted by grid map(which can be negative).	double
<code>maxh</code>	The height of the highest point cloud intercepted by grid map(which can be negative).	double
<code>leaf_size</code>	The resolution of point cloud. The higher it is, the faster the grid-based map is generated and the sparser the generated map.	double

The program will intercept the point clouds with height within the range of `[minh, maxh]` to compress into a 2D grid-based map. Observe the grid-based map in RViz, and if the generated grid-based map is incomplete, increase the height range , that is, `[minh, maxh]`, of the intercepted point cloud; if the generated grid-based map has too many black shadow areas, reduce the height range of the intercepted point cloud. The larger the height range of the intercepted point cloud, the longer it takes to generate the grid-based map. After modifying the parameters, save the profile and re-execute the grid-based map

command in the terminal to generate a new grid-based map.

[Notes] If the mapping still fails, please contact the technical support personnel.

3.3.3 Indoor Multi-Floor Mapping

[Notes] Please start the mapping program while the robot stands still.

[Notes] The `start_mapping.sh` script for single-floor mapping is not used to start the multi-floor mapping, because this script will start to create the point cloud map and grid map together, while the multi-floor needs to start the two mapping programs separately.

1. Open the terminal and enter the following command to open the point cloud program:

```
1 | bash ~/jy_cog/slam/scripts/mapping.sh indoor true livox
```

[Notes] The parameter `true` at the end of the command indicates that RViz needs to be enabled. If RViz does not need to be enabled, it can be changed to `false`.

[Notes] The point cloud map program started by this command must be kept until all floors are completed. Do not start and stop this program at the same time as the following grid map program starts and stops.

2. Please keep the robot standing still for 5~8 seconds after starting the point cloud program.
3. Whenever you arrive a new floor, you need to open the program to generate the grid map and name the map:

```
1 | bash ~/jy_cog/slam/scripts/occupancy_mapping.sh gridmapName
```

A parameter is carried at the end of the command, separated by a space from the command, indicating the file name of the grid map that is being generated for this new floor. When you need to generate multiple grid maps, you need to specify a different file name for each grid map.

[Notes] It is recommended to name the map *jueying*, which is the startup layer for navigation because the grid map with the name *jueying* is loaded by default when the navigation starts.

[Notes] The grid map can only be generated when the point cloud program is running.

- When the program to generate the grid map is running, user can control the robot to scan the environment of the floor. When finishing scanning this floor and arriving at the next floor, press **Ctrl+C** to quit the program of generating the grid map, and the grid map of this floor will be automatically saved. Then open the grid map program again in place (note that the map file name is specified and is not repeated), you can start to scan the next floor.

[Notes] It is recommended to close and restart the grid map program in the area connecting the two floors and ensure that the area is accessible on both two grid maps.

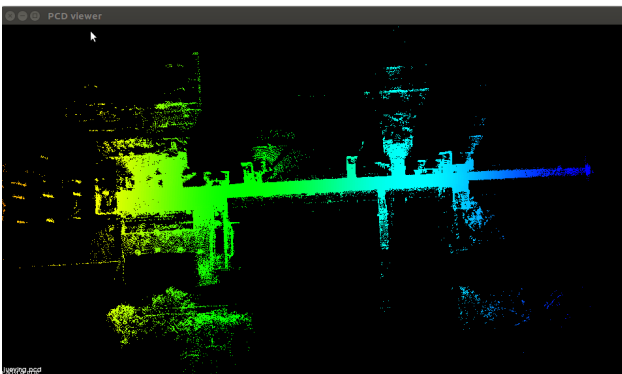
- After scanning all floors, press **Ctrl+C** to exit the grid map program and the point cloud map program, and both the grid map of the last floor and the complete point cloud map are automatically saved.

3.3.4 View Point Cloud Map

Open the terminal and enter the following commands to open PCD viewer to view the created 3D point cloud map:

```
1 cd ~/jy_cog/system/map
2 pcl_viewer jueying.pcd
```

PCD viewer is as shown in the figure below.

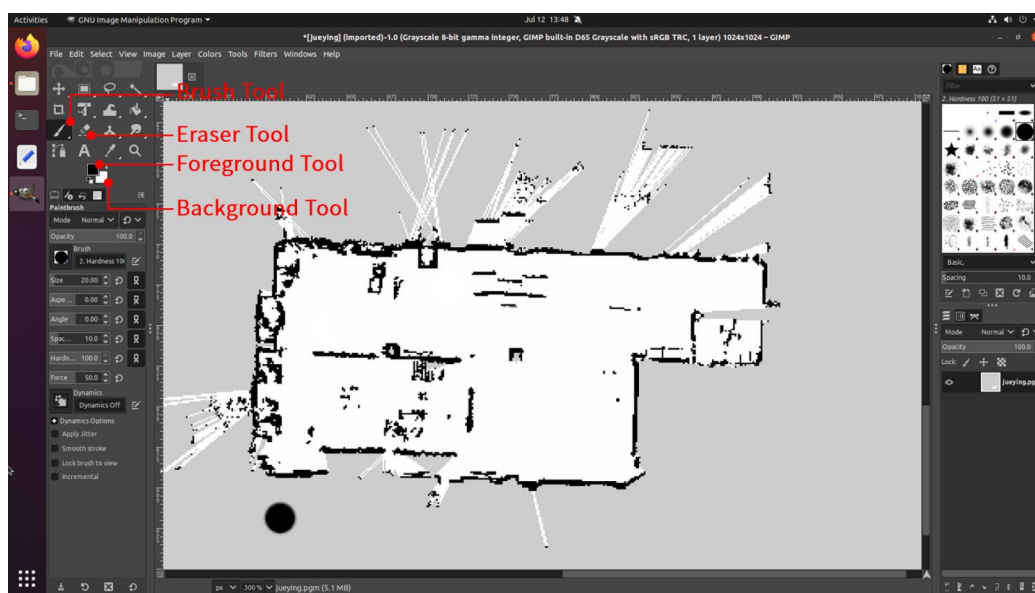
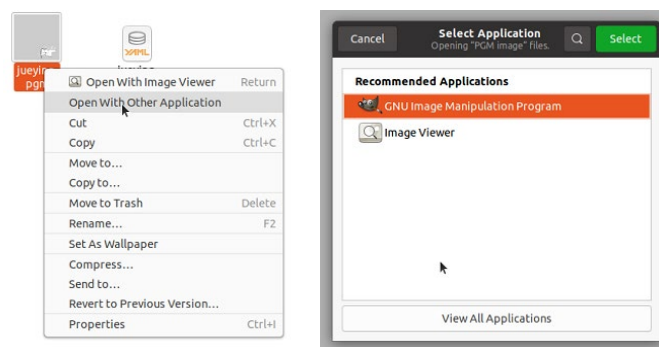


Basic operations of PCD viewer:

- Scroll up or down to control the zoom of the point cloud model.
- Drag the left button to control the rotation of the point cloud model.
- Drag the pulley to control the translation of the point cloud model.
- Use the numeric keys (1-5) to select the display mode of the point cloud model.

3.3.5 Modification of Grid Map

If the grid-based map does not fully match the actual environment, or it needs to manually define the passable areas, you can use GNU Image Manipulation Program to modify the .pgm file: Right-click the .pgm file, select Open with Other Application, then select GNU Image Manipulation Program, and click Select.

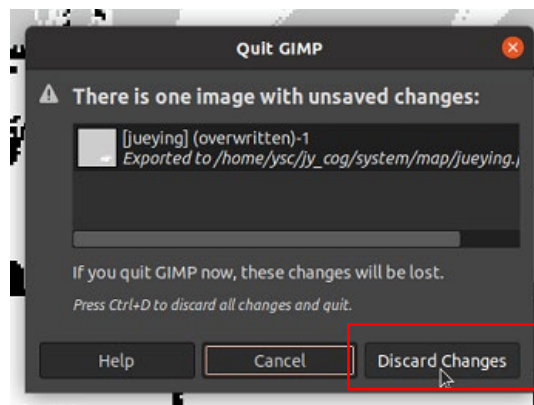


If the toolbars of GNU Image Manipulation Program are not displayed, click Tab key to display them.

The toolbars provide the tools of foreground color and background color, foreground color sets the color of the Brush tool, and background color sets the color of the Eraser tool. In the grid-based map, the black area represents the obstacle, that is, the impassable area. The white area represents the passable area, and the gray area represents the unknown area. You can use the Brush tool or the Eraser tool to add or erase the black areas. Press and hold the Shift key, then left-click to draw the straight line, or press and hold the Ctrl key, then left-click to suck up the color.

[Notes] If the map is multi-floor, it is important to ensure that the transition area between the two connecting floors is accessible from both two grid maps. If stairs or slopes are present in the map, they may be identified as obstacles and displayed as black in the map. When editing the map, the accessible area of the stairs or slope should be modified to white.

After the modification, select File - Overwrite jueying_map.pgm in the upper menu bar to overwrite the original file (do not choose File - Save to save the file). When the software is closed, the save prompt will pop up again. At this time, just click Discard Changes.



3.3.6 Map Configuration

For multi-floor mapping, after editing the grid map, you need to modify the configuration file *map.toml* in *~/jy_cog/system/conf/broker* (in newer versions, the folder *broker* has been renamed *robot_server*): Add or subtract the map in the map list according to the grid map of each floor. The serial number will be used to identify the grid map of each floor when the navigation task is issued.

```

1 Size = 1
2
3 [0] ← the serial number of this grid map
4 map_path = "/home/ysc/jy_cog/system/map/" ← the path of this grid map
5 map_prefix = "jueying" ← the file name of this grid map
6 map_suffix = [".pgm", ".yaml", ".png"]
7
8 [1]
9 map_path = "/home/ysc/jy_cog/system/map/"
10 map_prefix = "jueying1"
11 map_suffix = [".pgm", ".yaml", ".png"]

```

3.4 Map Deployment (for v2.1.0 and later)

This version's mapping function will simultaneously construct point clouds and 2D grid maps. For multi-floor scenarios that require multiple grid maps, you need to specify them manually when the floors change. The mapping program in this version supports map construction for single-floor small indoor scenes (50m×50m), multi-floor small indoor scenes (50m×50m×10 floors), and single-floor large scenes (500m×500m).

3.4.1 Notice for Use

1. During mapping, please do not manipulate the robot for strenuous exercise and do not block the laser radar, otherwise the mapping may fail.
2. Before mapping, open the terminal and enter the `top` command to check the

computing resources. If any other program uses too many computing resources, enter the following command line to close it: `sudo kill [PID]`.

```

ysc@ysc-perception:~$ top
top - 10:43:20 up 6 min, 1 user, load average: 1.26, 0.81, 0.37
Tasks: 283 total, 2 running, 281 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.3 us, 4.0 sy, 0.0 ni, 89.1 id, 0.1 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 15872.0 total, 13393.9 free, 1389.6 used, 1088.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 14008.3 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 2175 ysc        20   0 436292 26328 10404 R   27.9   0.2   1:41.68 yesense+
 2127 ysc        20   0 518480 29676 13056 S   19.3   0.2   1:16.53 rslidar+

```

3. Make sure the mapping program is not running. If it is running, please save the map and exit the program first according to 3.4.3. Make sure the navigation program is not running. If it is running, please enter the following command line in the terminal to kill it:

```
1 | bash ~/jy_cog/system/scripts/kill_nav.sh
```

4. Currently, the mapping for mixed indoor and outdoor scenes, smooth tunnels, smooth long corridors, relatively open scenes (depending on the radar model), high dynamic environments, extreme vibrations or vigorous movements, and long-distance scenarios without loops (or with few loops) venue is not supported.

3.4.2 Function Introduction

This version of the mapping program includes three mapping modes:

1. **fast:** Quickly construct point clouds and grid maps (including multi-floor grid map construction, but without loop detection and keyframe map generation).
2. **indoor:** Construct point clouds and grid maps for indoor multi-floor, narrow, or small-scale scenarios (including loop detection and keyframe map generation).
3. **outdoor:** Construct point clouds and grid maps for large, relatively open outdoor

scenarios (including loop detection and keyframe map generation).

The mapping program in this version will create a new file for each 3D point cloud map corresponding to a scene under the `/home/ysc/jy_cog/system/maps` directory. The folder is named using the format `[CustomName]-Date-Time`, and it contains a complete point cloud map (`jueying.pcd`), the default grid map (`jueying.pgm`), and a parameter file (`jueying.yaml`). If there are multiple grid maps for the same scene, they will be named with the prefix `jueying`, followed by a sequential number starting from 1, such as `jueying1` (while still retaining `jueying.pgm` and `jueying.yaml`). Additionally, keyframe maps (including point clouds of keyframes and their pose files) may be stored in the details folder.

The localization and navigation program will by default call the map located in the `/home/ysc/jy_cog/system/map` directory. The map folder is a symbolic link pointing to a specific map folder. If you need to change the address that the symbolic link points to, you can delete the original map folder and then follow the command below in the terminal to proceed:

```
1 cd ~/jy_cog/system          #enter the directory (optional)
2 ln -s maps/[folder name] map # Replace the [folder name] according to the actual situation. This
  command creates a symbolic link named map, pointing to a specific folder under the maps directory.
```

3.4.3 Single-Floor Mapping

[Notes] This version of mapping program is suitable for the indoor scenes with an area not more than $50\text{m} \times 50\text{m}$ and large scenes with an area not more than $500\text{m} \times 500\text{m}$.

[Notes] Please start the mapping program when the robot is standing.

1. Open the terminal and enter the following command to start the indoor mapping script:


```
1 bash /home/ysc/jy_cog/system/scripts/start_mapping.sh [MapName] [ImmediateEffect] [MappingMode]
[OpenRViz]
```

In this command line, four parameters can be appended at the end:

Parameter	Function	Value&Meaning
MapName	Specify the [CustomName] part of the map folder name, the default is "ProjectName-LocationName".	/
ImmediateEffect	Specify whether to immediately use this map for localization and navigation after the mapping is completed. If set to take effect immediately, the map symbolic link will point to this map folder. The default is to take effect immediately.	y = take effect immediately n = not take effect immediately
MappingMode	Specify the mapping mode to be used this time; the default is the "fast" mode.	fast/indoor/outdoor
OpenRViz	Specify whether to open the RViz interface when starting the mapping; the default is to open it.	true=open RViz false= not open RViz

For example:

```
1 # Example 1
2 bash /home/ysc/jy_cog/system/scripts/start_mapping.sh substationB y outdoor true
3 # Example 1 Execution Result: Mapping is performed using the "outdoor" mode, with the mapping
  results stored in the "substationB-Date-Time" folder under the /maps directory. After mapping,
  the results are immediately used for localization and navigation, and RViz is opened during the
  mapping process.
4 # Example 2
```

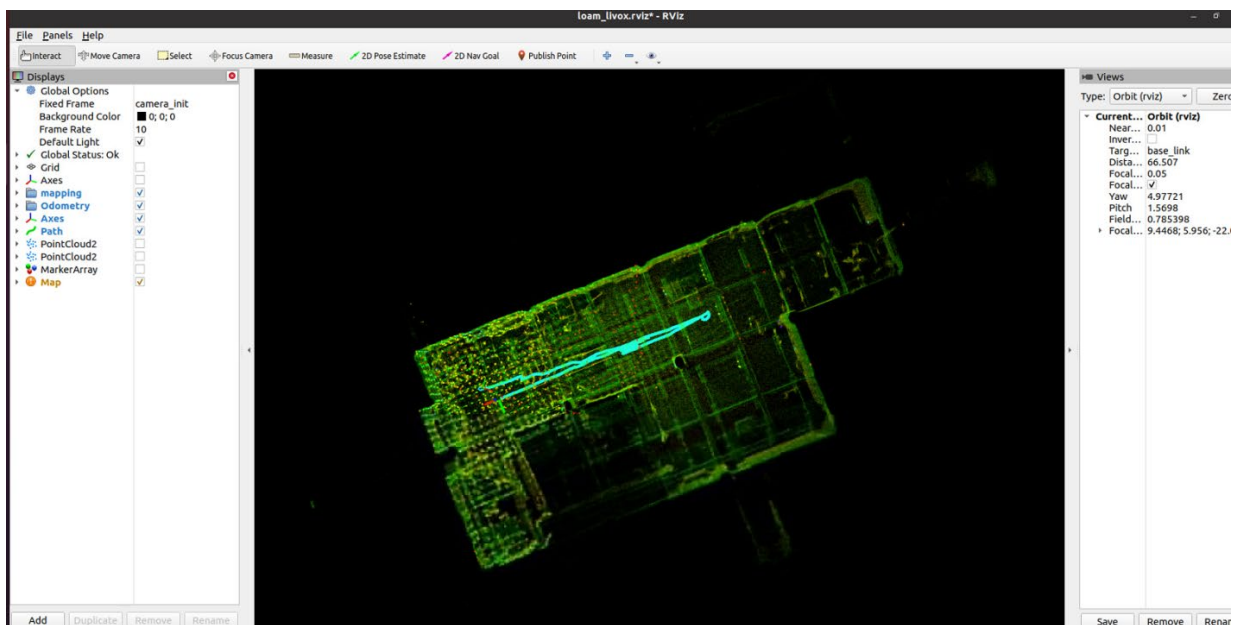
```

5  bash /home/ysc/jy_cog/system/scripts/start_mapping.sh
6  # Example 2 Execution Result: If the parameters at the end are omitted, the mapping will be
   performed using the default "fast" mode. The mapping results will be stored in the "ProjectName-
   LocationName-Date-Time" folder under the /maps directory. After mapping, the results will be
   used for localization and navigation by default, and RViz will be opened during the mapping
   process by default.

```

[Notes] The parameters at the end can only be omitted starting from the last one; you cannot skip a parameter in the middle. The program will read the parameters in sequence.

- After starting the mapping program, please first keep the robot standing still for 10 seconds, then remotely control the robot to walk at constant speed smoothly and steadily in the area that needs to be mapped. During scanning, you can check whether the localization of robot is normal and whether the point cloud is drifting in RViz.



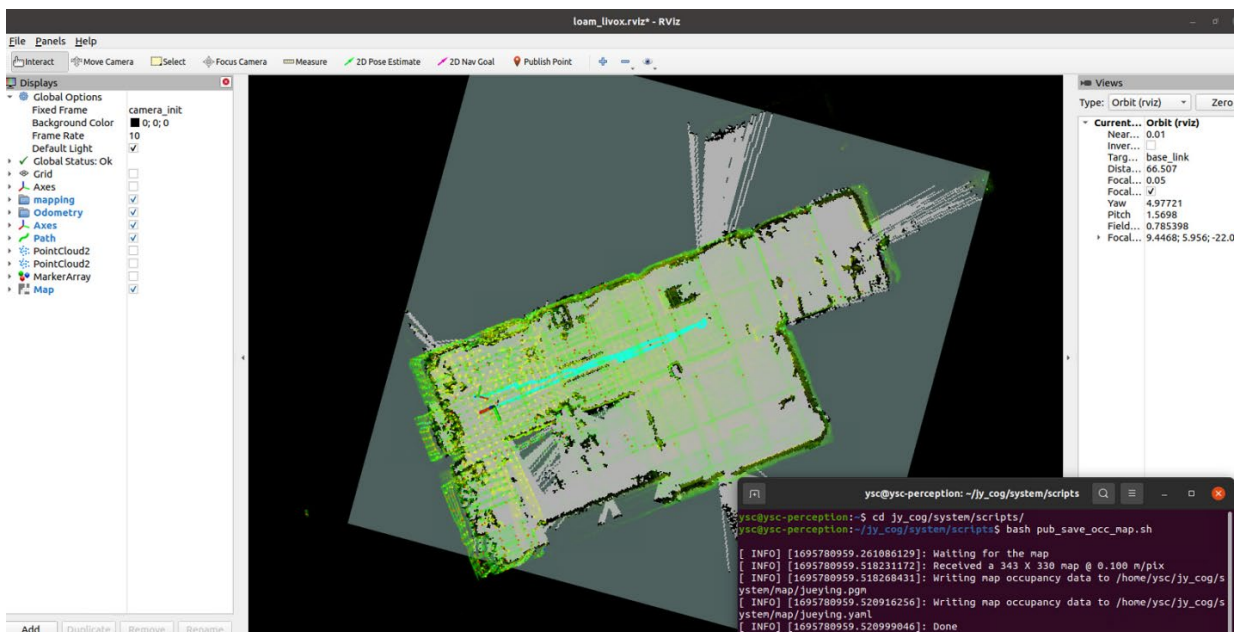
- After the robot finishes the scanning of the mapping area, press **Ctrl+C** in the terminal which starts mapping to quit mapping. If the fast mode is used during mapping, the program will automatically generate a grid map and save it to the corresponding map folder. If the indoor/outdoor mode is used, only several keyframe point clouds and their corresponding poses are saved in the map folder. You need to enter the following command lines in the terminal to separately start the programs for constructing the

complete point cloud map and the grid map:

- 1 `bash /home/ysc/jy_cog/system/scripts/construct_full_map.sh` # Program for constructing the complete point cloud map
- 2 `bash /home/ysc/jy_cog/system/scripts/construct_occ_map.sh` # Program for constructing the grid map

4. In fast mode, if you want to preview the grid map during the scanning process (not supported in indoor/outdoor mode), you can enter the following command in a new terminal:

- 1 `cd ~/jy_cog/system/scripts`
- 2 `bash pub_save_occ_map.sh`



[Notes] If the mapping program cannot be closed, please run the following command in the terminal to kill the mapping process:

- 1 `cd ~/jy_cog/system/scripts`
- 2 `bash kill_mapping.sh`

3.4.4 Multi-Floor Mapping

3.4.4.1 fast mode

[Notes] Please start the mapping program while the robot stands still.

[Notes] Multi-floor mapping will generate multiple grid maps corresponding to different floors based on the point cloud data of each floor. While controlling the robot to scan the environment, as

you switch floors, you need to use the `save_and_restart_occ_mapping.sh` script to convert the scanned floor's point cloud data into a grid map and restart the grid mapping program.

1. Open the terminal and enter the following command to start the point cloud mapping program:

```
1 | bash /home/ysc/jy_cog/system/scripts/start_mapping.sh [MapName] [ImmediateEffect] [MappingMode]
   | [OpenRViz] # For an explanation of the parameters at the end, please refer to section 3.4.3.
```

[Notes] This command starts the point cloud mapping program, which must remain running until the mapping of all floors is complete. Do not start and stop the process initiated by this command along with the restarts of the grid mapping program that follow.

2. After starting the mapping program, please keep the robot standing still for 10 seconds, and then remotely control the robot to walk smoothly and at a steady speed in the area where mapping is required.
3. Each time you start scanning a new floor, you need to manually run the following command line at the starting position of that floor (you need to terminate the process first by pressing `Ctrl+C` if the `save_and_restart_occ_mapping.sh` script is already running in the terminal) to save the grid map of the previous floor and restart the grid mapping program to begin constructing a new grid map:

```
1 | bash /home/ysc/jy_cog/system/scripts/save_and_restart_occ_mapping.sh
```

The grid map and parameter file for the initial floor where mapping starts are saved by default as *jueying.pgm* and *jueying.yaml*. Each subsequent time the `save_and_restart_occ_mapping.sh` script is run, the new grid map will be named with the prefix *jueying* followed by a sequential number, starting from 1, such as *jueying1.pgm*.

[Notes] The grid map can only be constructed when the point cloud mapping program is already running.

[Notes] It is recommended to stop and restart the grid mapping program on the stair landing connecting two floors, and ensure that the stair landing is accessible on both grid maps.

- After restarting the grid mapping program, you can remotely control the robot to scan the environment of the new floor. Once all scanning is complete, press **Ctrl+C** to stop the mapping program, and it will automatically save the point cloud and grid maps.

[Notes] Press **Ctrl+C** in the terminal only once; pressing it multiple times will send a signal to kill the program, which may prevent the map from being saved.

3.4.4.2 indoor/outdoor mode

[Notes] Please start the mapping program while the robot stands still.

[Notes] Multi-floor mapping will generate multiple grid maps corresponding to different floors based on the point cloud data of each floor. While controlling the robot dog to scan the environment, as you switch floors, you need to use the *save_and_restart_occ_mapping.sh* script to convert the scanned floor's point cloud data into a grid map and restart the grid mapping program.

- Open the terminal and enter the following command to start the point cloud mapping program:

```
1 bash /home/ysc/jy_cog/system/scripts/start_mapping.sh [MapName] [ImmediateEffect] [MappingMode]
   [OpenRViz] #For an explanation of the parameters at the end, please refer to section 3.4.3
```

[Notes] This command starts the point cloud mapping program, which must remain running until the mapping of all floors is complete.

- After starting the mapping program, please keep the robot standing still for 10 seconds, and then remotely control the robot to walk smoothly and at a steady speed in the area where mapping is required.
- When constructing multi-floor grid maps in indoor/outdoor mode, you need to specify the new floor's number when reach the starting position of the new floor:

```
1 bash /home/ysc/jy_cog/system/scripts/set_floor_label.sh [FloorNumber]
```

The above command includes a parameter [FloorNumber], with the following meanings for the parameter:

- 1) Omitting this parameter: This indicates using the default floor number value of 0.
- 2) Values such as 0, 1, 2, etc: Specify the new floor's number.
- 3) Value of -1: Indicates that no grid map will be constructed for this new floor.
- 4) Value of -2: Indicates that the new floor's number will be incremented by 1 based on the previous floor's number.

The resulting grid map will be named "jueying+FloorNumber," such as *jueying1.pgm*.

The grid map with a floor number of 0 will be named *jueying.pgm*.

[Notes] The floor numbers in the map files do not necessarily correspond one-to-one with the actual floor numbers.

[Notes] It is recommended to name the grid map file of the floor where the robot's navigation is initiated as "jueying," since the navigation system will, by default, load the map named "jueying" at startup.

[Notes] The grid map can only be constructed when the point cloud mapping program is already running.

[Notes] It is recommended to stop and restart the grid mapping program on the stair landing connecting two floors, and ensure that the landing is accessible on both grid maps.

4. After specifying the new floor number, you can remotely control the robot to scan the environment of the new floor. Once all scanning is complete, press **Ctrl+C** in the terminal where the mapping was initiated, to stop the mapping program. At this point, only several keyframe point clouds and their corresponding poses are saved in the map folder. You will still need to separately start the program to construct the complete point cloud map and the grid map:

```
1 | bash /home/ysc/jy_cog/system/scripts/construct_full_map.sh # Program for constructing the complete
```

```
point cloud map
2 | bash /home/ysc/jy_cog/system/scripts/construct_occ_map.sh # Program for constructing the grid map
```

3.4.5 View Point Cloud Map

Users can navigate to the target map folder they wish to view and use the PCD viewer to display the constructed 3D point cloud map, following the steps outlined in section 3.3.4.

3.4.6 Modification of Grid Map

Users can navigate to the target map folder they wish to view and follow the steps outlined in section 3.3.5 to modify the grid map.

3.4.7 Map Configuration

For multi-floor mapping, after constructing the grid map, you need to go to the `~/jy_cog/system/conf` directory and enter the `/broker` or `/robot_server` folder to modify the *map.toml* configuration file. Please refer to section 3.3.6 for detailed steps.

3.4.8 Record Data Packet

For larger scenes, this version of the mapping program cannot guarantee that the map will be completed in a single attempt. It is necessary to record data packets and repeatedly adjust parameters and reconstruct the map to achieve the best quality.

1. Start recording the data packet:

```
1 | bash ~/jy_cog/rosbag_recorder/scripts/start_record.sh
```

After executing the above command, wait approximately 5 seconds for the frequency check. If the frequency is normal, recording will automatically start (as shown in the left image below). If there is an issue with the frequency, recording will not begin (as

shown in the right image below, a [Warn] for sensor data and an [Error] are printed).

```

ysc@ysc-perception:3.0:~$ bash ~/jy_cog/rosbag_recorder/scripts/start_record.sh
ysc@ysc-perception:3.0:~$ [Info] 使用模式：mapping-LIO 进行录制。
[Info] 录制话题： ['/livox/lidar', '/imu/data', '/leg_odom']
[Info] 正在检查话题频率，请等待 4 秒。
[Warn] ERROR: Cannot load message class for [livox_ros_driver2/CustomMsg]. Are y
our messages built?
[Info] 话题： ['/imu/data'] 的频率是 199.977. 期望频率是： 200.0 Hz
[Info] 话题： ['/imu/data'] 的频率正常。
[Info] 话题： ['/leg_odom'] 的频率是 199.963. 期望频率是： 200.0 Hz
[Info] 话题： ['/leg_odom'] 的频率正常。
[Info] 通过此命令录制： rosbag record /livox/lidar /imu/data /leg_odom -o /home/y
sc/jy_cog/system/record/bags/mapping/LIO --tcpnodelay
[INFO] [1722922913.311445918]: Subscribing to /imu/data
[INFO] [1722922913.314894932]: Subscribing to /leg_odom
[INFO] [1722922913.316771706]: Subscribing to /livox/lidar
[INFO] [1722922913.331699561]: Recording to '/home/ysc/jy_cog/system/record/bag
s/mapping/LIO_2024-08-06-13-41-53.bag'.

ysc@ysc-perception:3.0:~$ bash ~/jy_cog/rosbag_recorder/scripts/start_record.sh
ysc@ysc-perception:3.0:~$ [Info] 使用模式：mapping-LIO 进行录制。
[Info] 录制话题： ['/livox/lidar', '/imu/data', '/leg_odom']
[Info] 正在检查话题频率，请等待 4 秒。
[Warn] ERROR: Cannot load message class for [livox_ros_driver2/CustomMsg]. Are y
our messages built?
[Info] 话题： ['/leg_odom'] 的频率是 200.011. 期望频率是： 200.0 Hz
[Info] 话题： ['/leg_odom'] 的频率正常。
[Warn] No new message of [/imu/data].
[Error] 某些话题频率异常，停止录制！

```

- Control the robot to scan the recording area. After completing the scan, execute the following command to stop the recording:

```
1 | bash ~/jy_cog/rosbag_recorder/scripts/stop_record.sh
```

The recorded point cloud data is stored in the `~/jy_cog/system/record/bags/mapping` directory.

[Notes] If you need to perform multi-floor mapping, please pause for a while at the initial position of each new floor to allow sufficient time for running the floor-splitting script in the subsequent mapping steps.

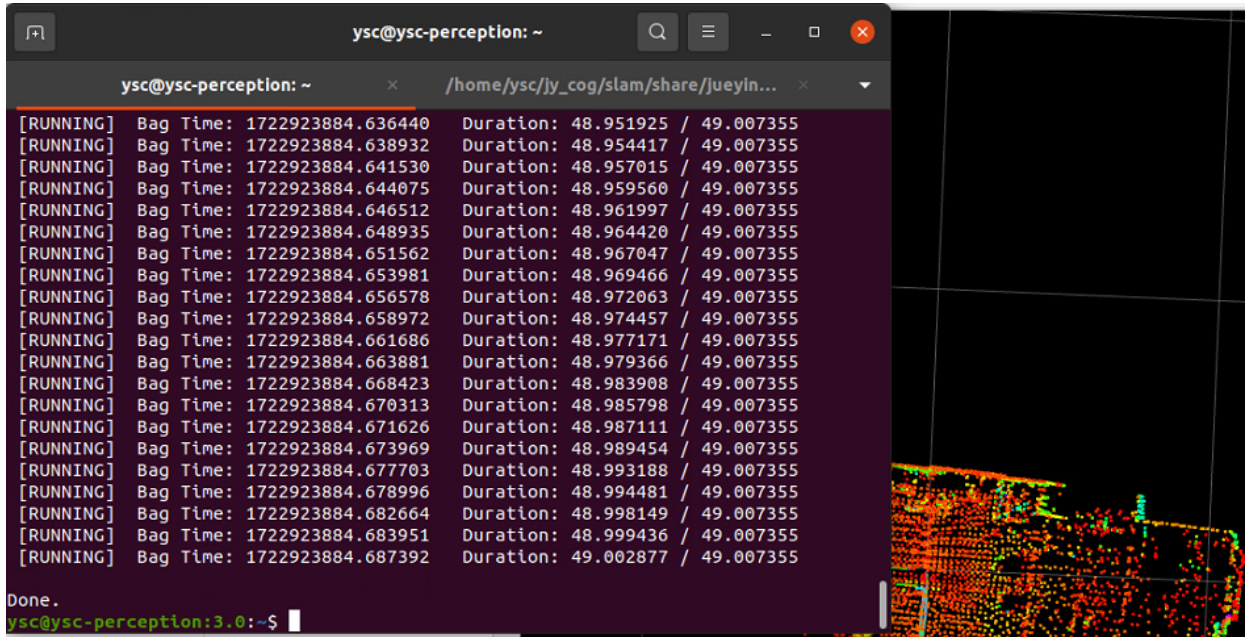
- Before using the recorded data packet for mapping, you need to kill the sensor driver:


```
1 | rostopic kill /livox_lidar_publisher2
2 | rostopic kill /yesense_imu_node
```
- Then, following the previously described mapping procedures, start the mapping program.
- After starting the mapping, open a new terminal, navigate to the data packet storage path, and play the data packet:

```
1 | rosbag play ~/jy_cog/system/record/bags/mapping/ DataPacketName
```

[Notes] If performing multi-floor mapping, you can run the `set_floor_label.sh` script when the data packet playback reaches the entrance of the new floor.

- After the data packet playback is complete (as shown in the figure below), you can follow the previously described mapping procedures to finish and save the map.



3.5 Single floor navigation

Jueying X30 Pro adopts the ROS/navigation framework.

[Notes] The navigation methods described in this section and Section 3.6 should be used with localization($\geq 2.0.5$), nav($\geq 2.1.9$), player($\geq 1.0.6$), broker/robot_server($\geq 1.1.6$), and app($\geq 2.3.2$).

[Notes] When using the autonomous navigation function, you should avoid turning on sharp terrain, and pay attention to temporary obstacles that suddenly appear in the blind area of the robot's field of vision.

[Notes] Before using the navigation, ensure that the mapping program is not running.

[Notes] After finishing the map deployment, you can refer to 3.2 to change the contents of the file *autorun_slam* to true, and restart the robot, so that the localization and navigation program will start automatically after each startup.

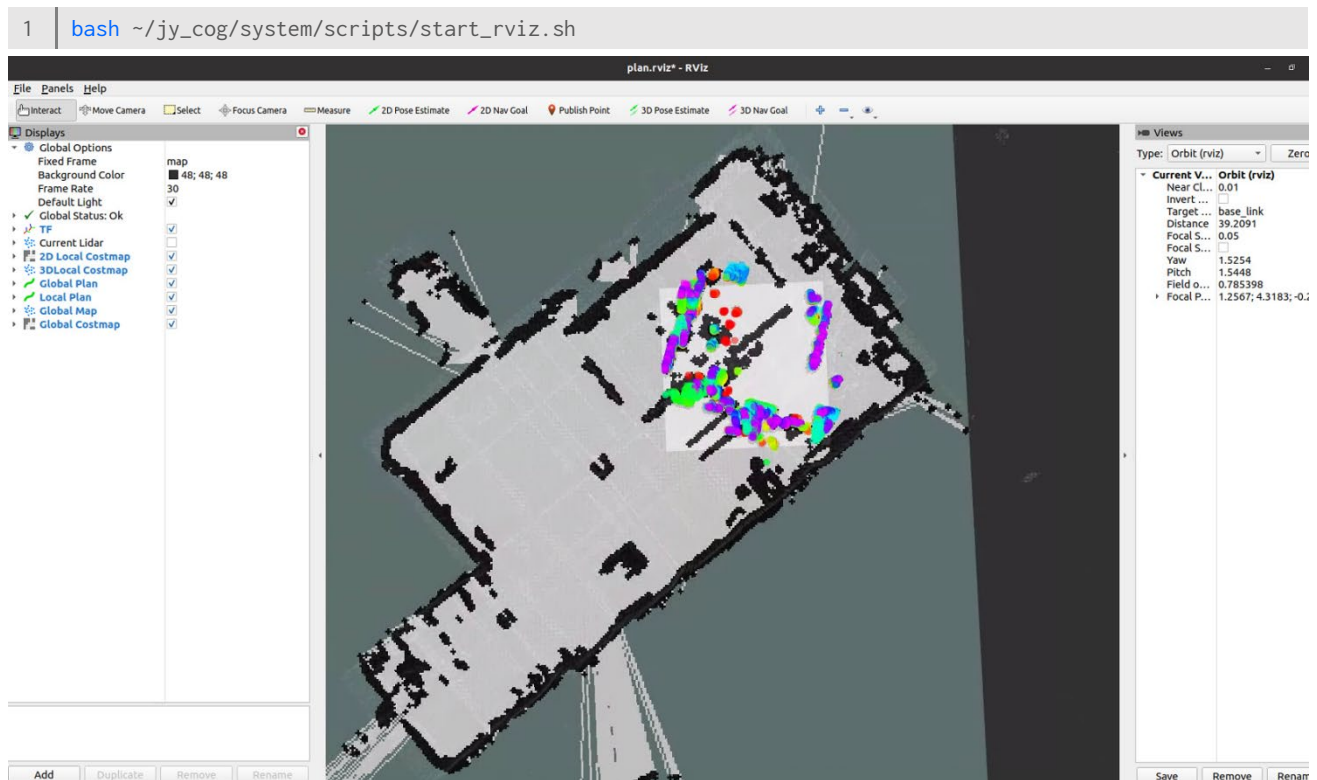
3.5.1 Start Localization and Navigation Program

Open the terminal and enter the following commands to start the localization and navigation program:

```
1 cd ~/jy_cog/system/scripts
2 bash start_nav.sh
```

After completing the above steps, RViz will pop up automatically, as shown in the figure below. RViz will display the map, and point cloud data collected by the robot in real time (in color).

If localization and navigation program is running (or set to be auto-started), you can open Rviz directly:



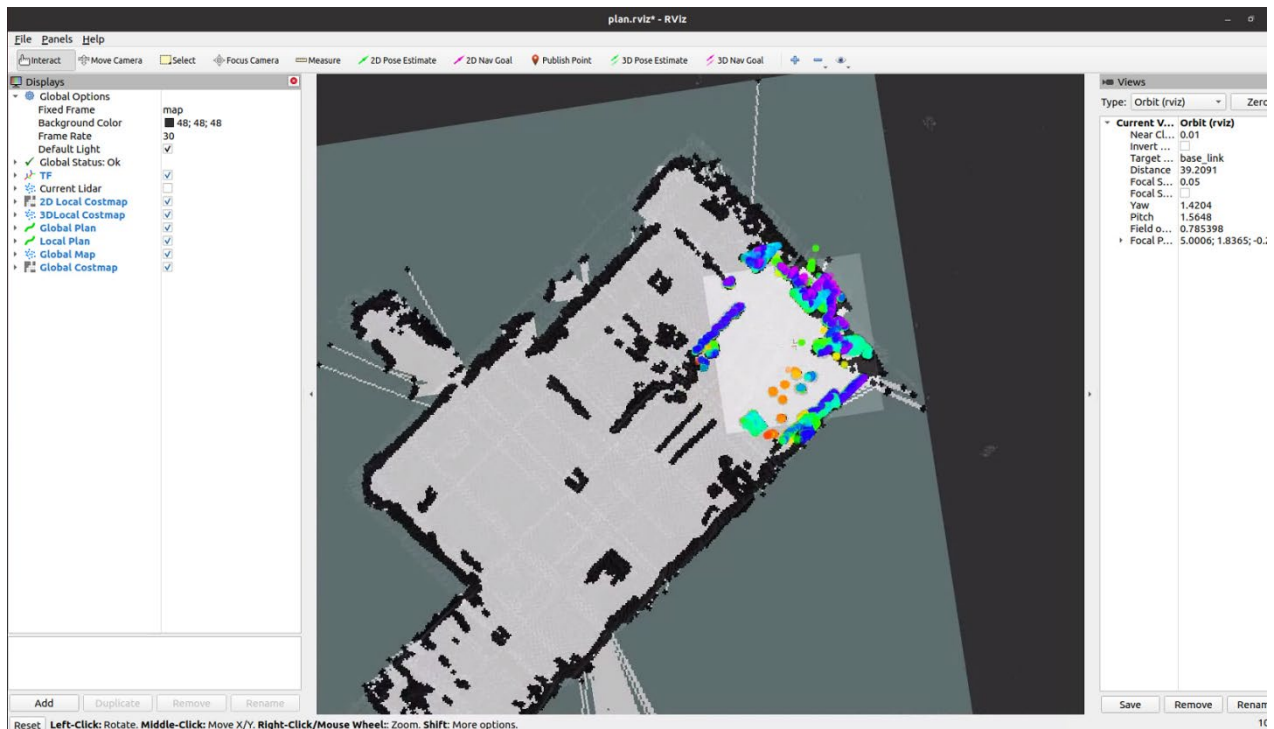
3.5.2 Initialize Localization

After starting the navigation, if the point cloud does not completely coincide with the grid-based map (as shown in the figure above), it needs to manually initialize the localization of robot.

1. Click [2D Pose Estimate] in the top toolbar.
2. Determine the actual position and orientation of the robot, move the mouse to the corresponding position in the map, press and hold the left mouse button and drag out

the arrow according to the actual orientation, then release the left mouse button.

3. If it succeeds to initialize the localization, the laser point cloud will coincide with the grid-based map, as shown in the figure below. If it fails to initialize the localization, just restart the operation from step 1.



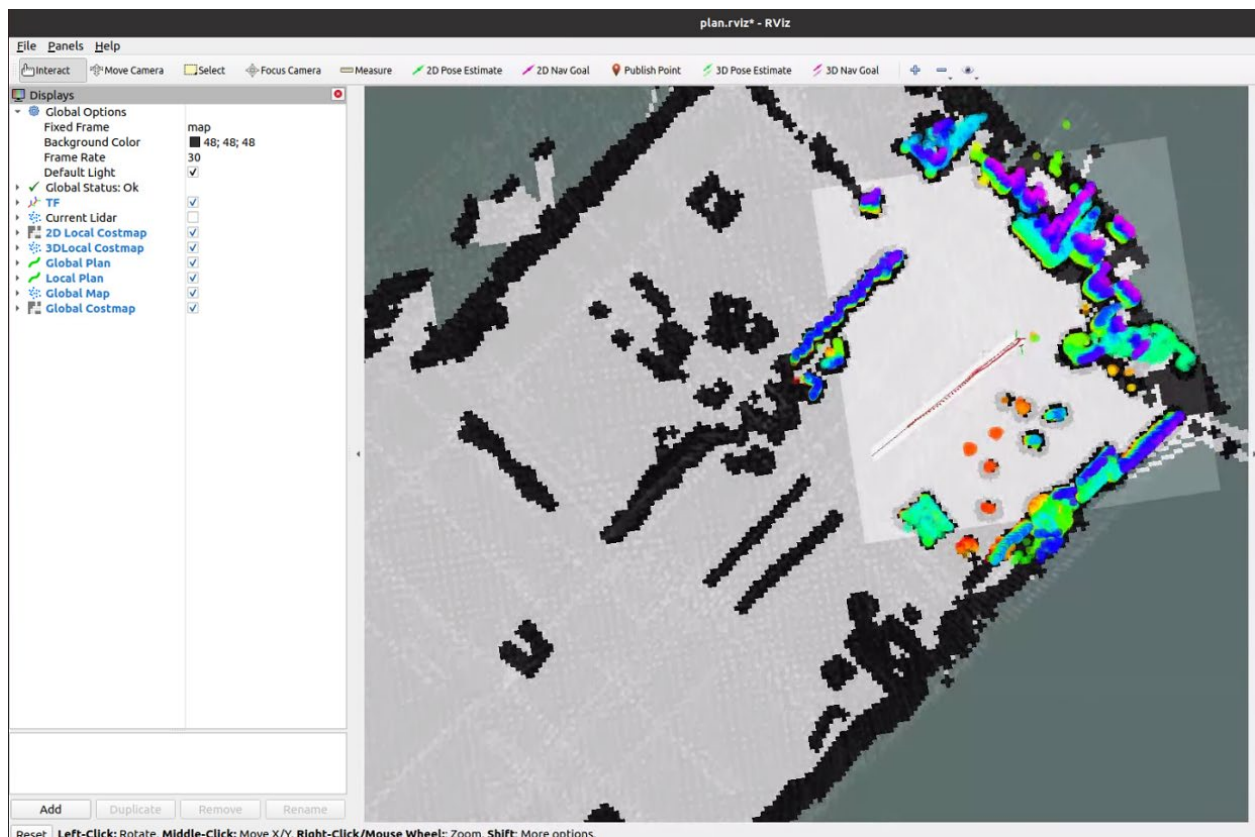
[Notes] If it is unable to locate, please refer to "3.5.6 Problem Identification".

3.5.3 Specify Target Point (Single Point)

After initializing the localization, you can specify a navigation target point for the robot on the map:

1. Click [2D Nav Goal] in the toolbar above.
2. Move the mouse to the target position in the map, press and hold the left mouse button and drag out the arrow according to the target orientation, then release the left mouse button.
3. If it succeeds to specify the target point, the planned motion path of the robot will

appear in the interface; If it fails to specify the target point, just restart the operation from step 1.



- Finally, switch the robot from manual mode to navigation mode on the controller, and operate the robot to start stepping, so as to make the robot to travel to the target point by itself. During navigation, the robot can autonomously avoid the obstacles outside a certain range.

[Notes] In autonomous navigation, all personnel must maintain a distance of more than 2m from the robot!

[Notes] If the robot is out of control during autonomous navigation, please switch it back from navigation mode to manual mode in time, or press the stop button (or toggle up SW1 and SW4 simultaneously).

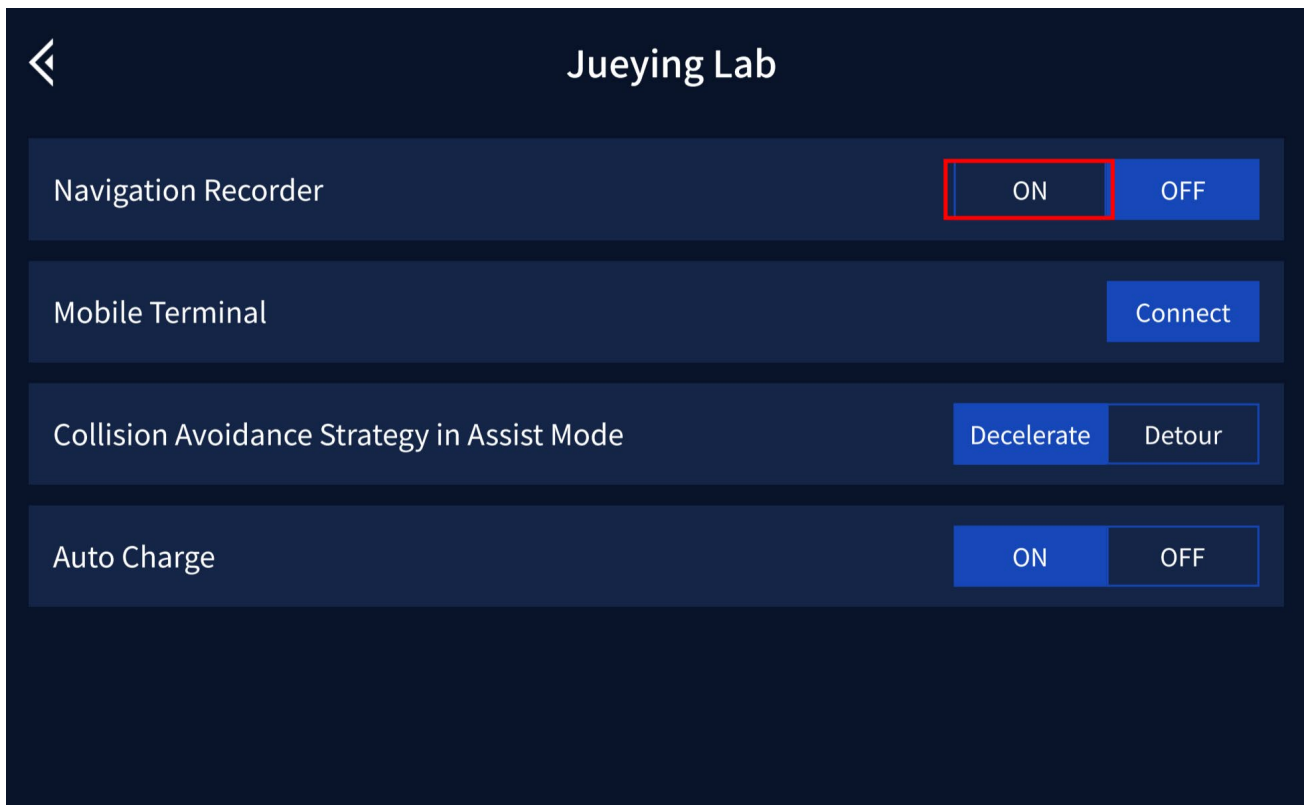
3.5.4 Usage of Navigation Recorder (Multi-Point Navigation)

The Controller APP provides a navigation recorder tool for recording navigation

waypoints and route, which enables the robot to reach a series of waypoints in sequence and complete tasks autonomously at the waypoints.

3.5.4.1 Preparations

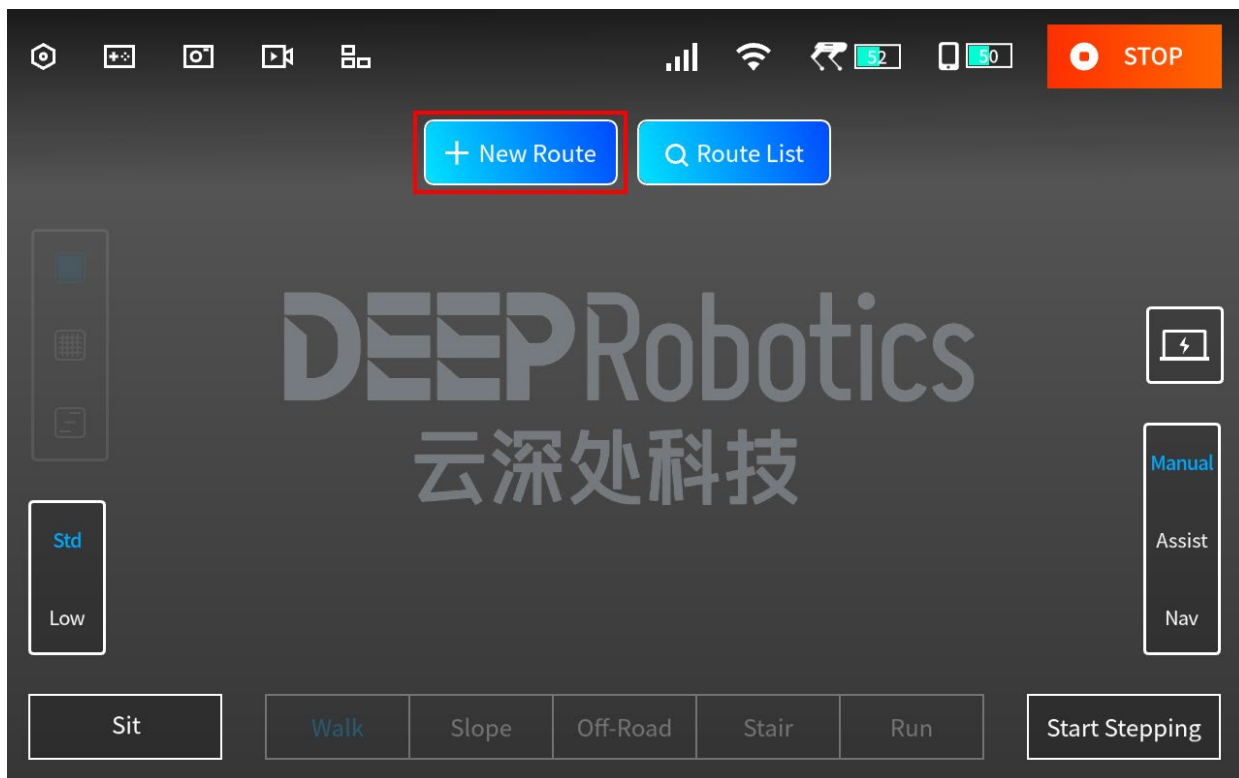
1. Open the Settings and enter Jueying Lab to activate Navigation Recorder.



2. Plan the path for the robot in advance, and allocate the locations of waypoints reasonably.

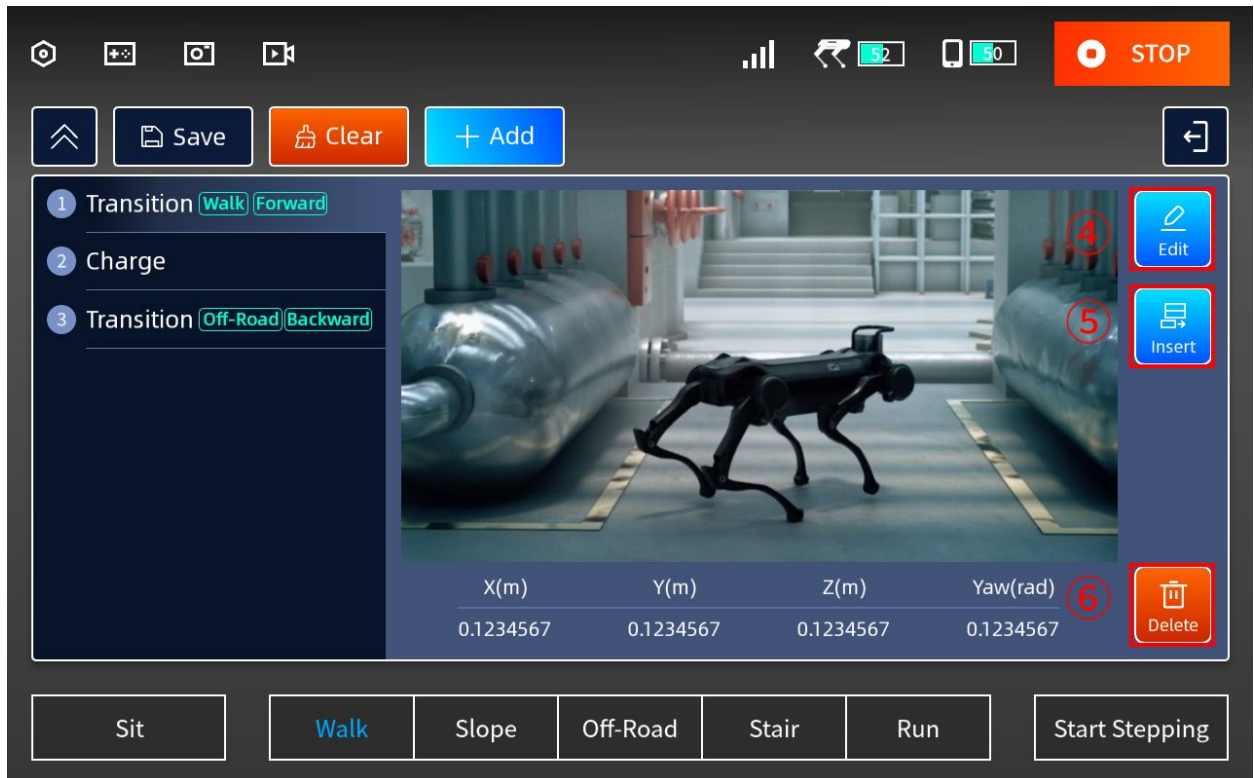
3.5.4.2 Create Navigation Route

1. Firstly, ensure that the robot is in manual mode or assist mode currently, and click the "+New Route" button on the interface to create a new route.



2. Then you can begin to record some waypoints of this route.





① Add	Add a new waypoint to the end of the waypoint list and set the waypoint properties
② Clear	Clear all waypoints in the current waypoint list
③ Expand/Collapse	Expand or collapse the waypoint list
④ Edit	Modifies the properties of currently selected waypoint (except location and orientation information)
⑤ Insert	Insert a new waypoint in front of the currently selected waypoint
⑥ Delete	Delete the currently selected waypoint from the list
⑦ Save Route	Save and name the current route and exit
⑧ Exit	Exit directly without saving the current route

3. Control the robot to the waypoint that needs to be recorded in sequence, then make

the robot stop stepping, click [①Add], and set the waypoint properties. The current location information of the robot is automatically obtained and user can update it.

⑨ Change MapID	Select the grid map where the waypoint is located
⑩ Update Location	Update the real-time robot location and orientation
⑪ Save Waypoint	Save the current waypoint and return
⑫ Waypoint Properties	Set waypoint properties. For details, see 3.5.4.3
⑬ Save the camera view	Select whether you want to save video stream photo taken at the current waypoint

4. After marking all waypoints, you can click the [⑦Save Route] button to save and name the route.

[Notes] When the navigation route includes climbing stairs, the waypoint before going upstairs must be at least 0.8m away from the stair edge.

[Notes] When add the entry of the Charge Waypoint, do not use L-Walk , Mountain or Silent Gait.

It's recommended to use Walk Gait or Slope Gait.

[Notes] When using a remote controller to deploy a navigation route, the distance between the first and last waypoint must not be too far away, or the navigation task cannot be cyclically executed.

3.5.4.3 Waypoint Properties

The properties of a waypoint are divided into **line properties** and **point properties**, where line properties represent the properties that take effect on the route to the waypoint.

Waypoint Type, Waypoint No., MapID, and Waypoint Location are point properties, while Body Height, Terrain, Direction, Gait, Speed, Navigation Mode, Collisions Avoidance Switch belong to path properties.

Waypoint Property		
MapID ^[1]	MapID is the serial number of grid map where the waypoint is located according to <i>map.toml</i> .	
Location	The location and orientation of waypoint in the map coordinate system.	
Waypoint Type	Transition	Transition point is used only to overcome terrains and constrain paths. Robot will keep stepping when it reaches a transition point and then move on to the next point.
	Task	When the robot arrives at the task point, it will make an accurate adjustment of the body, and then stop stepping, stay in place to perform inspection tasks such as meter identification, and then go to the next point.
	Charge	This type of waypoints should be recorded in front of the

Waypoint Property		
		charging dock so that the robot can locate the charging dock and enter to charge, and it will exit the charging dock immediately and go to the next point (if the charge point is the last point, the robot will keep charging)
Body Height	Standard	During the process of reaching this waypoint, the robot maintains its normal posture.
	Low	During the process of reaching this waypoint, the robot lowers its body (only supported when using walk or slope gait).
Terrain	Solid	On the way to this waypoint, the ground on which the robot places its feet is a solid surface, such as concrete floors or concrete stairs
	Grid	On the way to this waypoint, the ground on which the robot places its feet is a grid-like pattern with gaps in between, such as industrial stairs or platforms made of steel grating panels
	Composite	On the way to this waypoint, the ground on which the robot places its feet includes multiple types of ground surfaces. For example, the robot moves from solid ground to grid ground, or from grid ground to solid ground (collisions avoidance based on the local

Waypoint Property		
		elevation map is not supported on composite ground, and only stair gait and 45° stair gait are applicable on composite ground).
Direction	Move Forward	The direction of speed is the same as the body's orientation upon reaching this waypoint.
	Move Backward	The direction of speed is opposite to the body's orientation upon reaching this waypoint (used when going down the stairs).
Gait ^[2]	Walk	Use Walk gait to reach this waypoint
	Slope	Use Slope gait to reach this waypoint
	Off-Road	Use Off-Road gait to reach this waypoint
	Stair	Use Stair gait to reach this waypoint
	45° Stair	Use 45° Stair gait to reach this waypoint (only supported in Composite Mode)
	L-Walk	Use L-Walk gait to reach this waypoint
	Mountain	Use Mountain gait to reach this waypoint
Speed	Low	Reach this waypoint at low speed (not supported in Composite Mode)
	Medium	Reach this waypoint at medium speed (some gaits not supported)
	High	Reach this waypoint at high speed (some gaits not

Waypoint Property		
		supported)
Navigation Mode ^[3]	Straight	The route to this waypoint is planned globally in a straight line.
	Auto	Consider the terrain on the map when planning globally and calculate a route to bypass obstacles.
Collisions Avoidance Switch ^[4]	ON	Collision avoidance is enabled when navigation.
	OFF	Collision avoidance is disabled when navigation, to try to pass the narrow areas full of obstacles.

[1] MapID is used to identify the grid map of multi-floor navigation. For single floor navigation, MapID is generally 0 by default. If you want to check the MapID, go into `~/jy_cog/system/conf/broker` or `~/jy_cog/system/conf/robot_server`, and open the file *map.toml*, and view the serial number of the grid map (see 3.3.6).

[2] When using RL gaits (L-Walk and Mountain), body height and terrain adjustment are not supported.

[3] When using the Straight navigation, if two points cannot be planned globally in a straight line, the navigation task will fail. When using the Auto navigation, the path points should be set based on terrain complexity. The higher the terrain complexity, the higher the execution risk of the navigation path, and in this case, the more intensive target points should be set. Try to use multi-segment Straight navigation instead of Auto navigation.



[4] Whether the obstacle avoidance function of the navigation module is enabled or not, the obstacle avoidance function based on the local elevation map, as a safety module of the robot itself, will always remain enabled (Excluding composite ground sections, this obstacle avoidance function will remain disabled on composite ground segments). When the obstacle avoidance function of the navigation is enabled, for Straight navigation, the robot will **decelerate** and stop at an obstacle, and for Auto navigation, it will **detour** to avoid collisions.

3.5.5 Execution of Multi-point Navigation

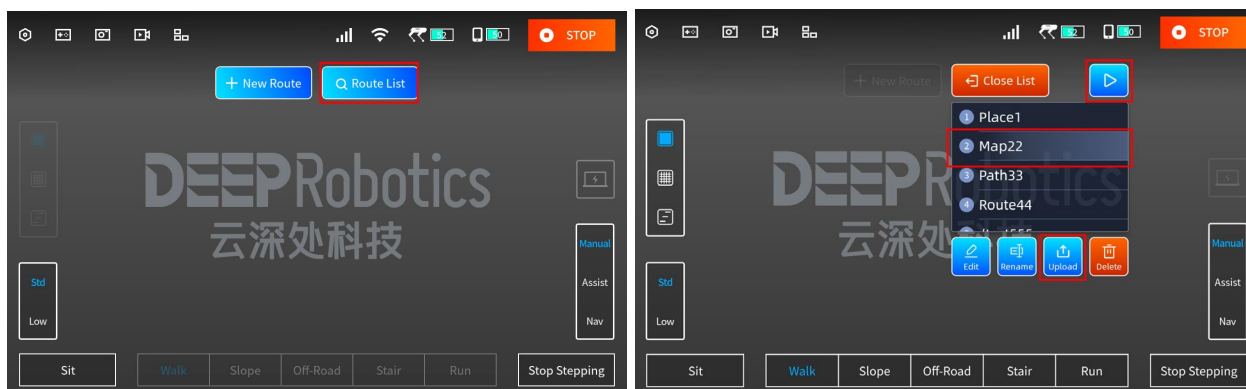
Multi-point navigation can be started in two ways: using remote controller or using player. The remote controller can only perform navigation tasks with a total number of points (number of waypoints included in the route $\times$ the number of times the route is looped) of no more than 100.

[Notes] Before performing navigation, ensure that the localization and navigation program is started and that the localization is initialized.

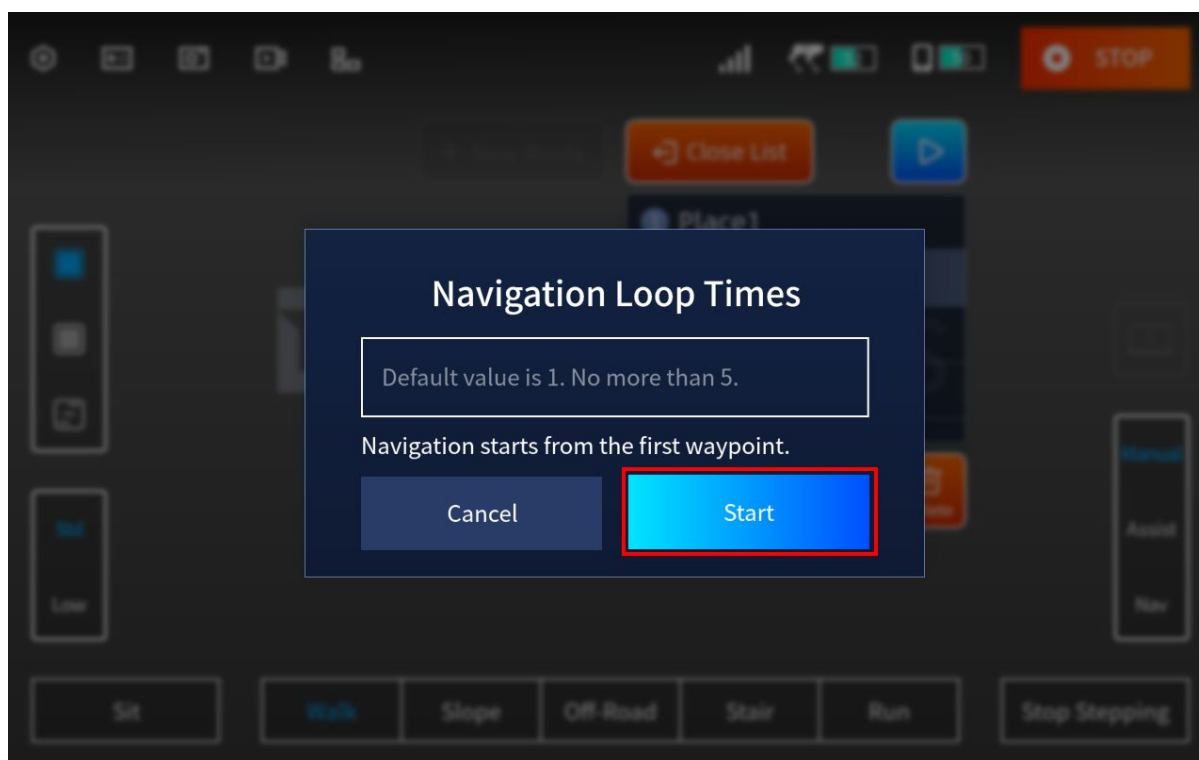
3.5.5.1 Using Controller for Navigation Execution

1. Click [Route List], select the navigation route to be executed from the list, click the

triangle button to execute the navigation route.

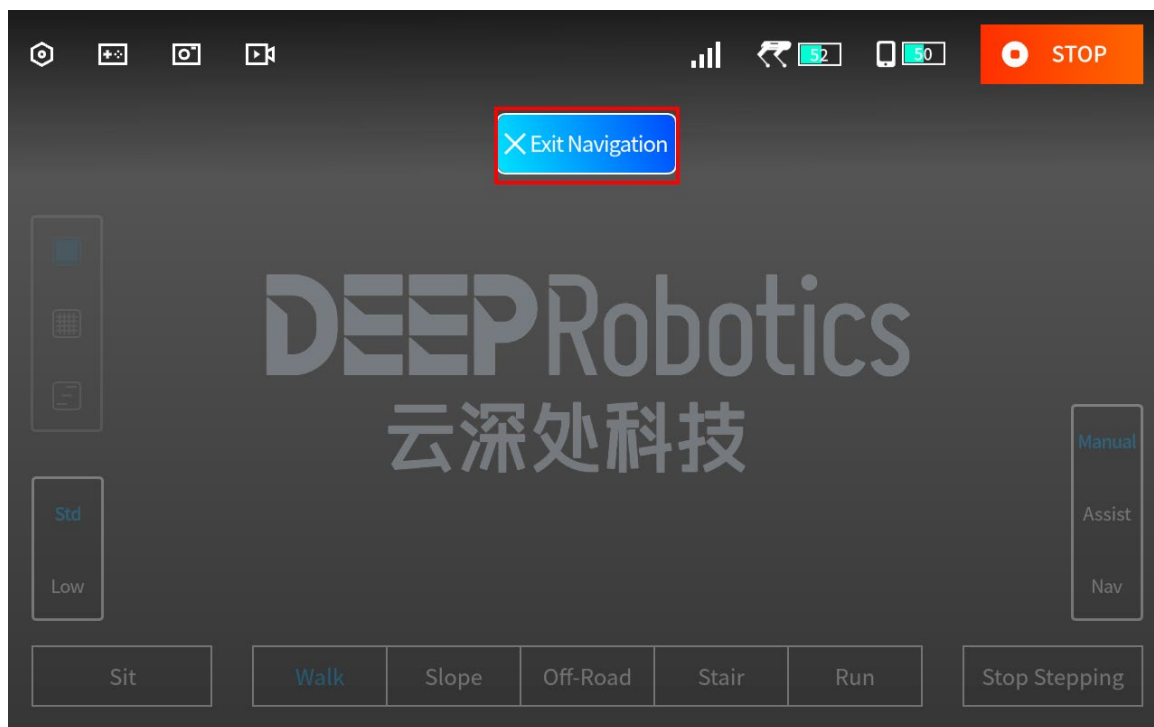


- Set the number of times the route is looped and click [Start].



[Notes] The total number of waypoints executed on the controller (the number of waypoints included in the route $\times$ navigation loop times) cannot exceed 100.

- During the execution of the navigation route, you can click [Exit Navigation] to exit the navigation.



3.5.5.2 Using player for Navigation Execution

If the total number of waypoints to execute exceeds 100, you can use **player** to perform.

1. Select the route you want to execute and click [Upload] to upload the route to the navigation host.



- Refer to 3.1 to connect to the navigation Host and go to the `~/jy_cog/system/data` folder, and check whether the route file (.json) is uploaded from controller.

[Notes] The name of the route file in navigation host is Controller ID+ route name.

- Go to the `~/jy_cog/system/conf/player` folder, open the `config.toml` file, and update the path of route file:

```

1 | points_path = "/home/ysc/jy_cog/system/data/routefile.json"    #the path of route file you want
   | executed
2 | End_wait_time = 1          #stop time after reaching the last waypoint(s)
3 | times = 3                 #route loop times
4 | charger_time = 500        #charging duration time of charge point(s)
5 | server_ip = "192.168.1.106"

```

- Modify other parameters in `config.toml` as needed.
- Save and close the modified `config.toml` file.
- Open a new terminal, enter the following command to run the player, and the robot will begin the navigation task:

```

1 | cd ~/jy_cog/player/scripts
2 | bash player.sh

```


7. To end the navigation, press ctrl+c in the terminal where the player is running.

3.5.6 Problem Identification

When there is no point cloud or the localization is not accurate, run the following command line to check whether the topic is normal:

Check the ROS topic of LiDAR data:

```
1 | rostopic hz /lidar_points
2 | #It is normal if the information can be printed and the average rate is close to 10Hz.
```

Check the ROS topic of IMU data:

```
1 | rostopic hz /imu/data
2 | #It is normal if the information can be printed and the average rate is close to 200Hz
```

Check the ROS topic of leg odometry data:

```
1 | rostopic hz /leg_odom
2 | #It is normal if the information can be printed and the average rate is close to 200Hz.
```

If all of the above information is normal but there are localization abnormalities, check whether the 3D point cloud map matches the actual environment by the instructions in "3.3.4 View Point Cloud Map". If not, please rebuild the map.

3.6 Indoor Multi-Floor Navigation

[Notes] Please read this section based on your understanding of single-floor navigation.

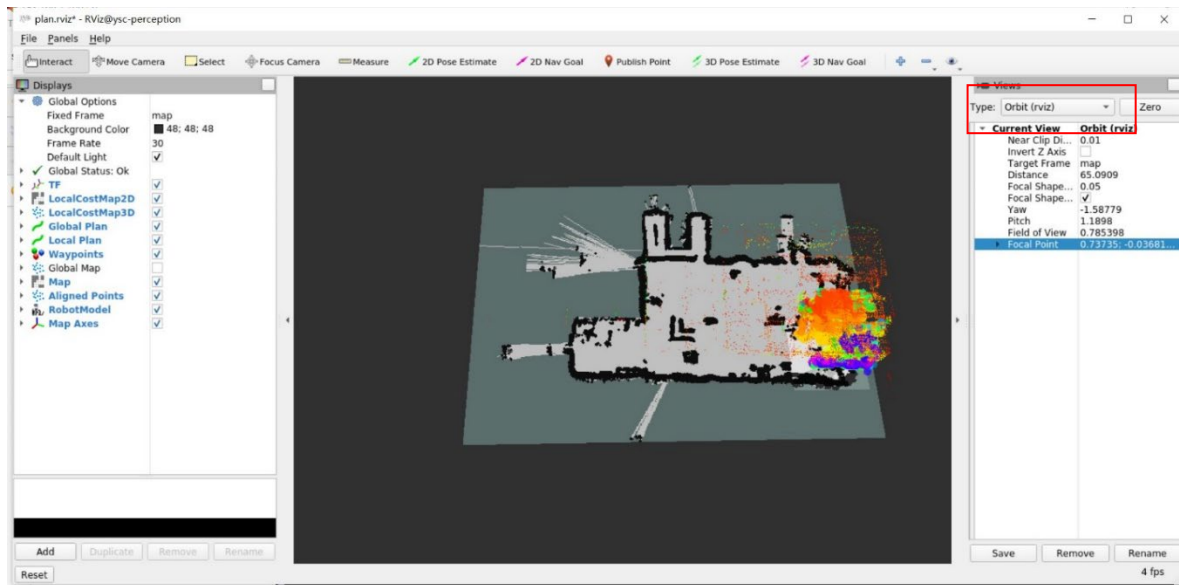
The main differences between multi-floor navigation and single-floor navigation are:

1. Multi-floor navigation requires initializing localization in the 3D point cloud map;
2. During multi-floor navigation deployment, you need to change the MapID to match the waypoint with the correct grid map.

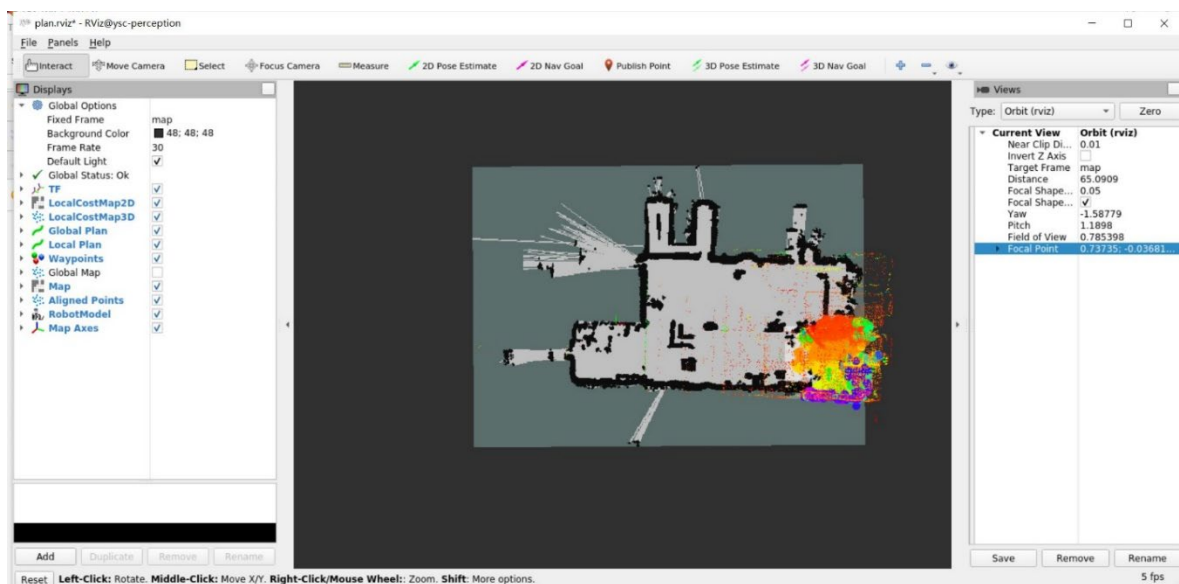
3.6.1 Localization Initialization

After navigation is started, if the point cloud does not fully coincide with the grid map, the robot location needs to be initialized.

1. Check that the Type under the Views bar on the right of the Rviz window is Orbit.



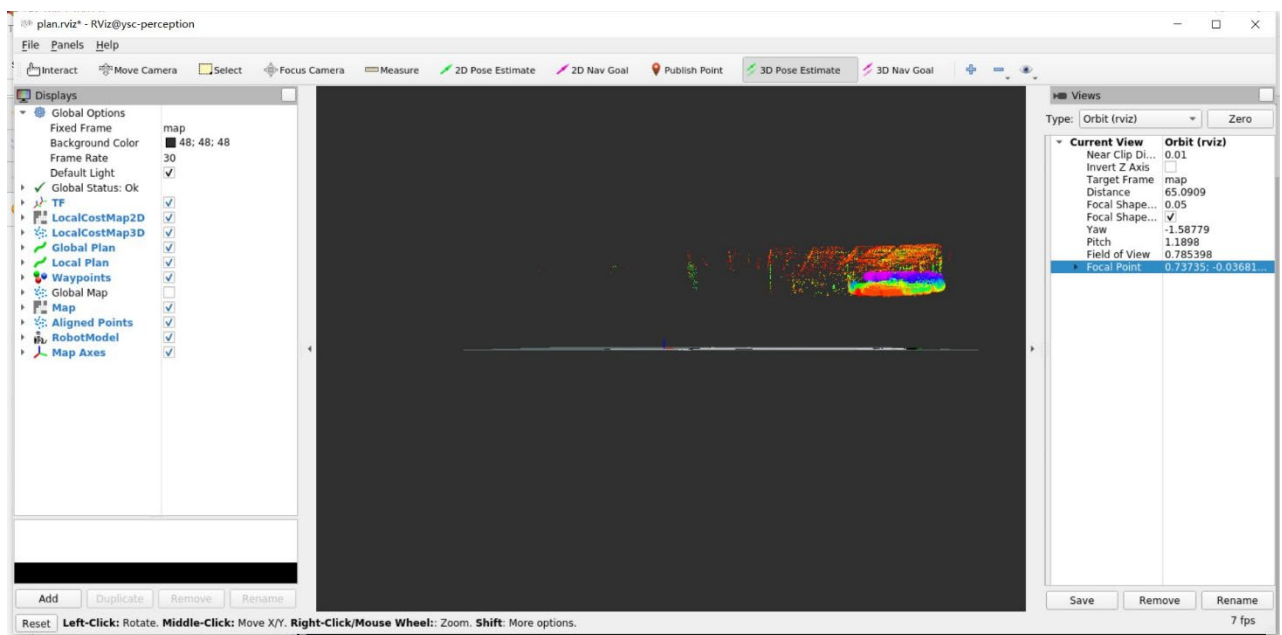
2. Adjust the perspective to a top-down view of the map.



3. Select the [3D Pose Estimate] tool on the upper toolbar.
4. Based on the grid map, find the corresponding position and orientation of the robot

on the grid map according to its position and orientation in the actual environment, move the mouse to the corresponding position in the map, press and hold the left mouse button and drag out the arrow according to the actual orientation, then release the left mouse button.

- Continue to adjust the perspective: press and hold the left mouse button to adjust the perspective to a horizontal view to observe the floor positions on the point cloud map.



[Notes] Do not use right mouse button to change the perspective. Press the right mouse button will change the robot location in the map.

- Press and hold the right mouse button to drag the robot position arrow to the height of the correct floor, release the right button, and the localization is initialized.

3.6.2 Grid Map Switching

When recording navigation waypoints, you need to click the button to change MapID and input the corresponding serial number in map.toml of the grid map where the new waypoints are located.

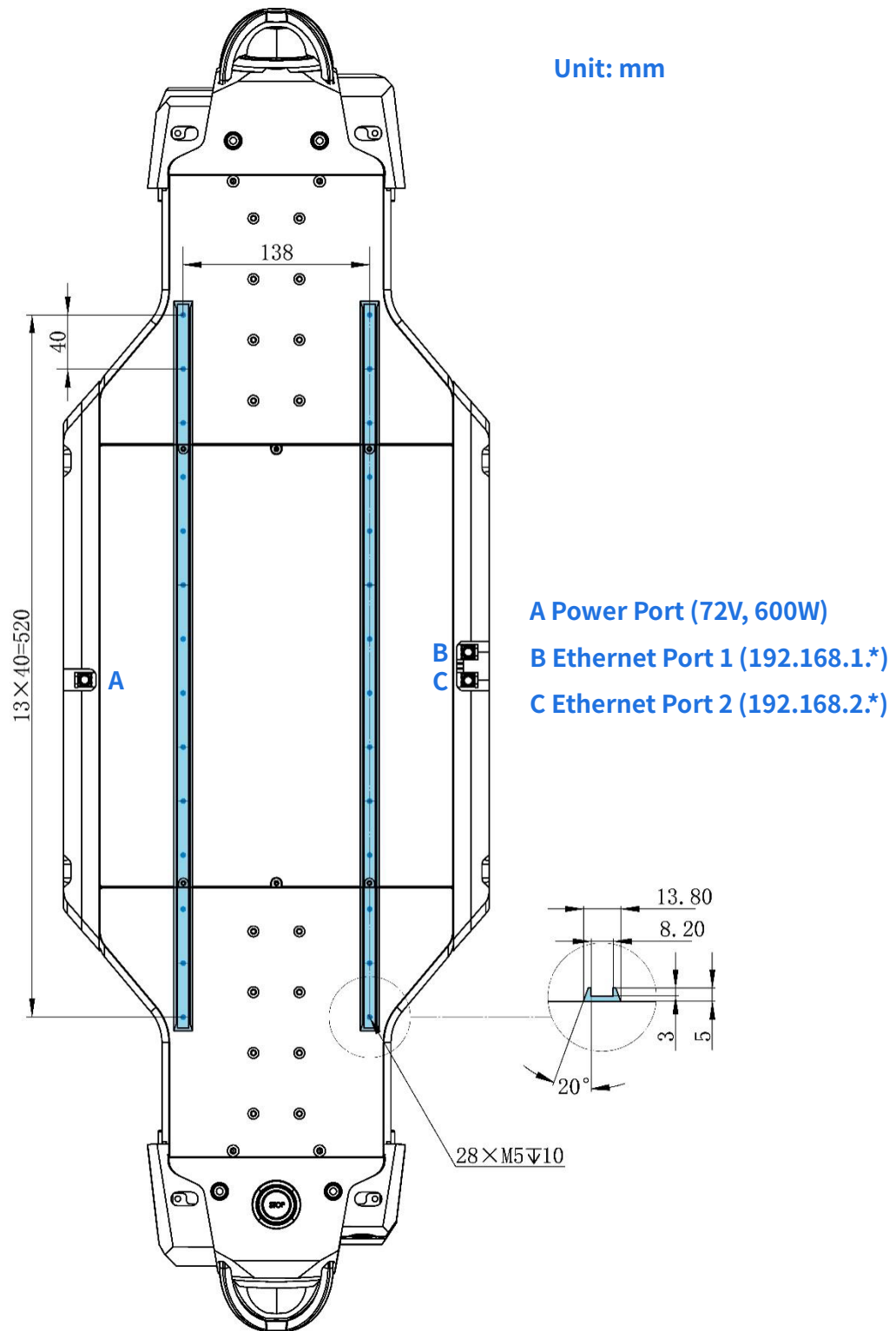
[Notes] When creating the first waypoint, make sure that the default MapID matches the actual grid map.

[Notes] When you reach the platform connecting the two floors and mark the waypoint of switching grid map, click the button to change MapID corresponding to the grid map of the next floor.

[Notes] Make sure that the waypoint of switching grid map is reachable on the both two grid maps.

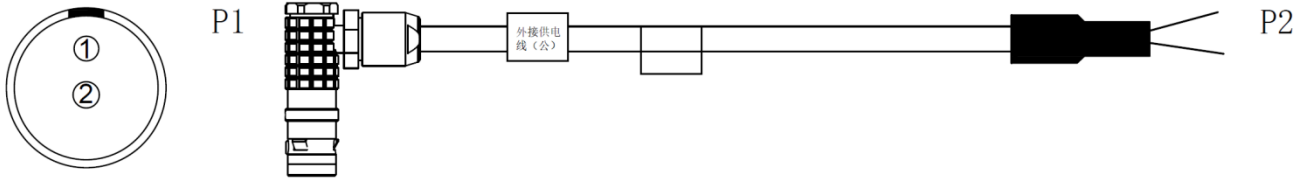
4 Payload Mounting

4.1 Mounting Holes and Hardware Interfaces



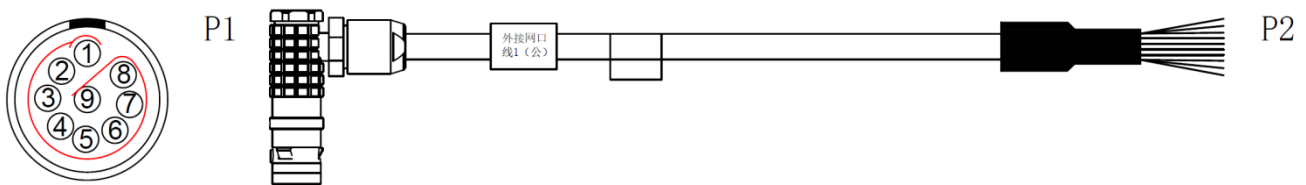
4.1.1 Power Port Wiring Color Code

[Notes] The power of this power port should not exceed 600W.



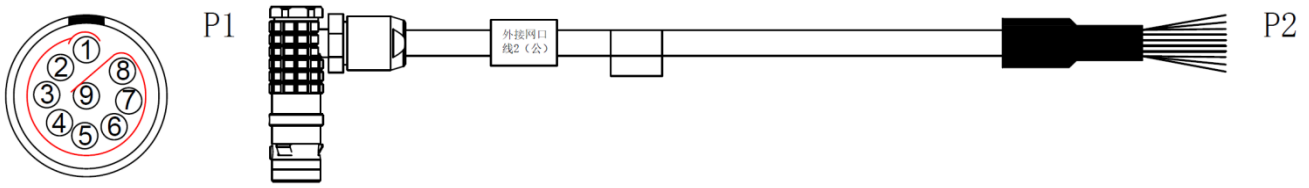
P1	Color	P2
1	Red	72V (+)
2	Black	GND

4.1.2 Ethernet Port 1 Wiring Color Code



P1	Color	P2
1	Orange White	TX+
2	Orange	TX-
3	Green White	RX+
4	Blue	TX2+
5	Blue White	TX2-
6	Green	RX-
7	Brown White	RX2+
8	Brown	RX2-

4.1.3 Ethernet Port 2 Wiring Color Code



P1	Color	P2
1	Orange White	TX+
2	Orange	TX-
3	Green White	RX+
4	Blue	TX2+
5	Blue White	TX2-
6	Green	RX-
7	Brown White	RX2+
8	Brown	RX2-

4.2 Mounting Limit

The load must be fixed on the holes of the two mounting rails, and the height of the load should not exceed 500mm.

[Notes] Please ensure that the load does not move, shake or rotate when fixing it.

[Notes] Please do not fix the load only on the holes of the same rail.

4.3 Motion Parameter Configuration

4.3.1 Payload Parameters

If the payload needs to be increased or decreased, the user needs to access the motion

host /home/ysc/jy_exe/conf, open the configuration file motion.toml, and configure the parameters for the payload weight and the position of the center of mass of payload. This can refer to 2.2 Parameter Configuration for operation. The new motion parameter configuration will take effect after the motion program is restarted.

4.3.2 Network Parameters

4.3.2.1 Add Load

If it needs to increase the load, and the new load needs to communicate with the motion host (for example, it needs to obtain the UDP messages reported by the motion host or issue UDP commands to the motion host), you need to configure the network parameters of the motion host after the load is installed.

1. Go to the /home/ysc/jy_exe/conf folder of motion host, and open the profile network.toml:

```
1 ip = '192.168.1.105'
2 target_port = 43897
3 local_port = 43893
4 ips = ["192.168.1.105", "192.168.1.106"]
5 ports = [43897, 43897]
```

2. In the square brackets of the last two line, add the IP address and port number for the new load receiving UDP messages according to the format, and the IP address and port number correspond one by one in order:

```
1 ip = '192.168.1.105'
2 target_port = 43897
3 local_port = 43893
4 ips = ["192.168.1.105", "192.168.1.106", "xx.xx.xx.xx"] # xx.xx.xx.xx is the IP address for the new
load receiving messages
5 ports = [43897, 43897, xxxxx] # xxxxx is the port number for the new load
receiving UDP messages
```


3. After configuring the network parameters, please restart the motion program, then you can communicate with the motion host according to the API Documentation. The IP address for the motion host receiving UDP messages is 192.168.1.103, and the port number is 43893.

4.3.2.2 Remove Load

If it needs to remove the load, and this load communicates with the motion host (for example, it needs to obtain the UDP messages reported by the motion host or issue UDP commands to the motion host), you need to configure the network parameters of the motion host after the load is removed.

1. Go to the `/home/ysc/jy_exe/conf` folder of motion host, and open the profile `network.toml`:

```
1 ip = '192.168.1.105'
2 target_port = 43897
3 local_port = 43893
4 ips = ["192.168.1.105", "xx.xx.xx.xx"]      # xx.xx.xx.xx is the IP address for the removed load
receiving UDP messages
5 ports = [43897,xxxxx]                    # xxxxx is the port number for the removed load receiving
UDP messages
```

2. Remove the IP address and port number corresponding to the removed load in the square brackets:

```
1 ip = '192.168.1.105'
2 target_port = 43897
3 local_port = 43893
4 ips = ["192.168.1.105"]
5 ports = [43897]
```

3. Save and close the file. New paramters will take effect after the motion program is restarted.

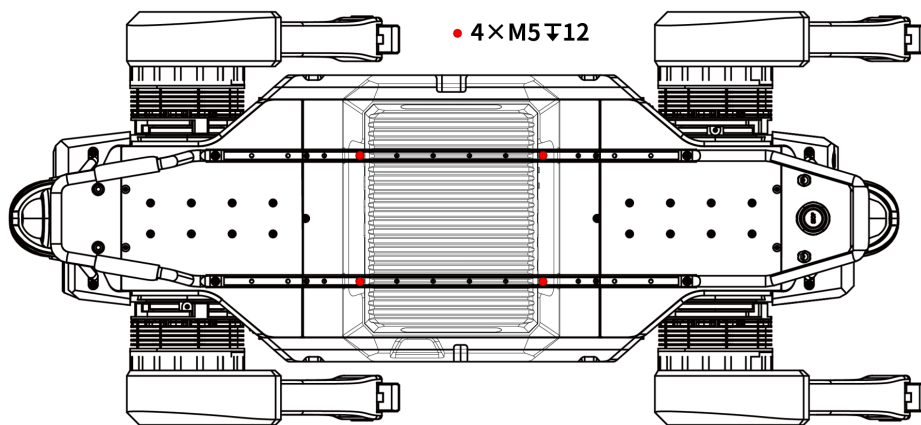
5 Intelligent Controller

5.1 Removal and Installation

5.1.1 Removal

If you want to remove intelligent controller:

1. First make sure that the robot is powered off.
2. Unscrew the four M5 screws from the mounting holes.



3. Pull out the three aviation plug from two sides of the robot with tool, and then remove the intelligent controller.
4. Start the robot and configure the motion parameters of the motion host according to 4.3 Motion Parameter Configuration . If the robot has no load, the following configuration parameters can be used:

a) network.toml:

```
1 ip = '192.168.1.105'
2 target_port = 43897
3 local_port = 43893
4 ips = ["192.168.1.105"]
5 ports = [43897]
```

b) motion.toml:

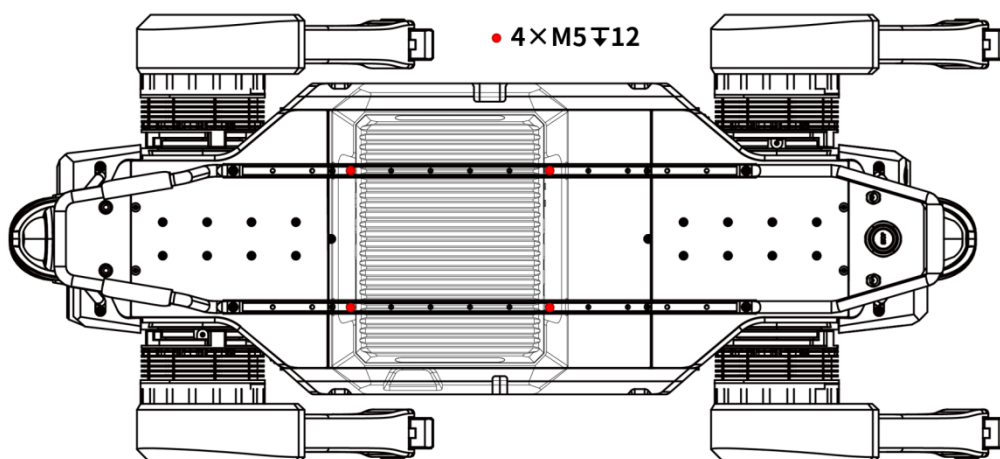
```
1 load = 0.0
2 .....
3 load_x = 0.0
4 load_y = 0.0
5 load_z = 0.0
```

5. Run the script `restart.sh` to restart motion program. Then the robot can be used normally.

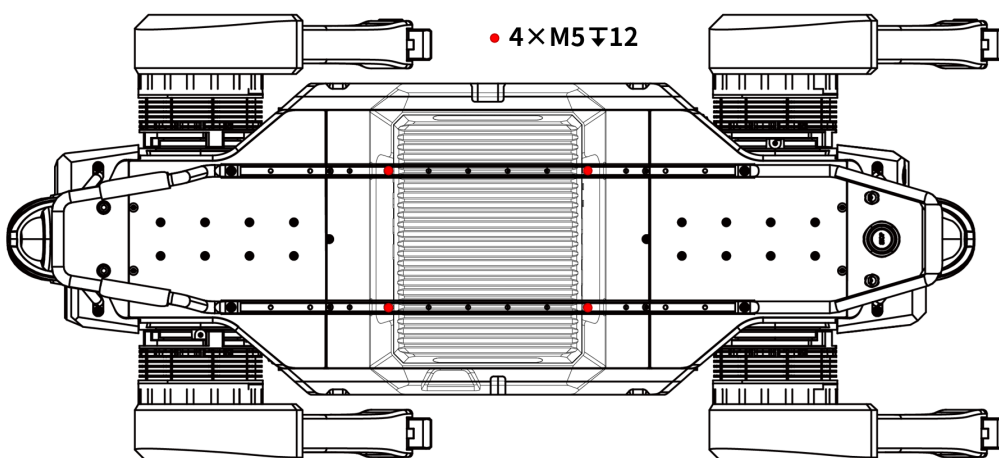
5.1.2 Installation

Intelligent controller can be installed in the following three positions to adjust the front and rear installation space and meet different installation requirement.

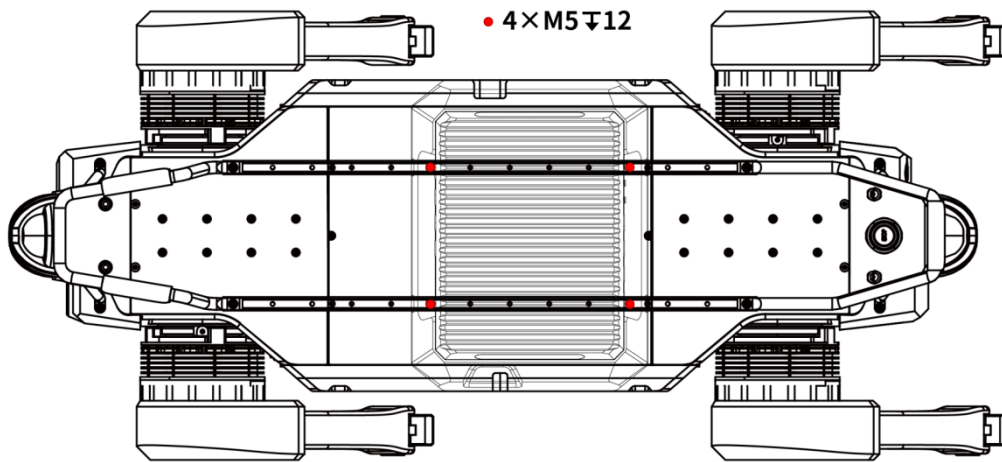
Position A:



Position B:



Position C:



Please follow the steps below for installation:

1. Make sure the robot is powered off.
2. Place the intelligent controller in a suitable position on the back of the robot. Insert the three aviation plugs on the intelligent controller into the corresponding aviation plug ports on both sides of the body, and tighten the four M5 screws to fix the intelligent controller on the two mounting rails.
3. Start the robot and configure the motion parameters according to the [4.3 Motion Parameter Configuration](#) . If the robot only has the intelligent controller installed on its back, you can use the following configuration parameters:

a) network.toml:

```

1 | ip = '192.168.1.105'
2 | target_port = 43897
3 | local_port = 43893
4 | ips = ["192.168.1.105", "192.168.1.106"]
5 | ports = [43897, 43897]
```

b) motion.toml:

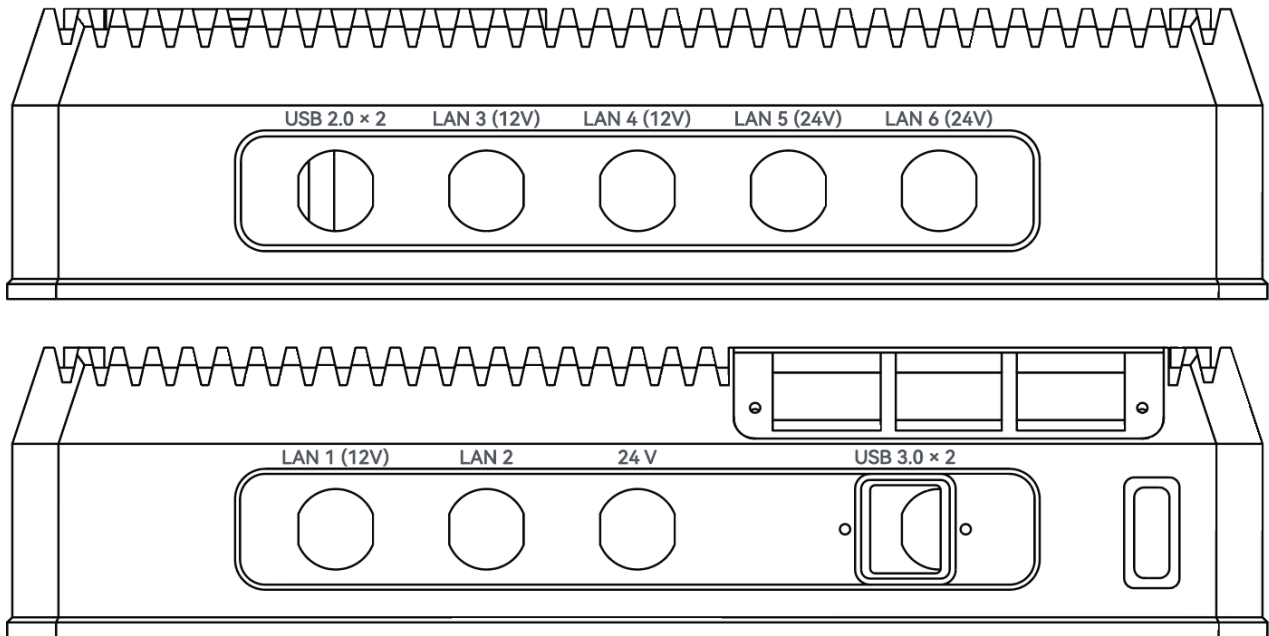
```

1 | load = 3.0
2 | .....
3 | load_x = 0.05
4 | load_y = 0.0
5 | load_z = 0.10
```

4. Run the script restart.sh to restart motion program. Then the robot can be used

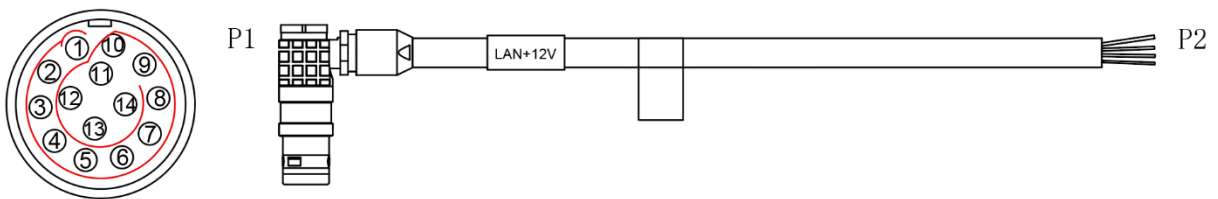
normally and the intelligent controller can be powered on.

5.2 Hardware Interface and Wiring



[Notes] The maximum power of LAN(12V) is 36W, and the maximum power of LAN(24V) is 72W and the maximum power of DC 24V is 480W. The total power supplied to the payload should not exceed 600W.

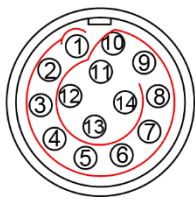
5.2.1 LAN1(12V) & LAN3(12V) & LAN4(12V)



P1	Color	P2
1	Red+Light Green	12V (+)
2		
3	Black+Pink	GND
4		

P1	Color	P2
5	Orange	TX+
6	Yellow	TX-
7	White	RX+
8	Blue	TX2+
9	Purple	TX2-
10	Green	RX-
11	Grey	RX2+
12	Brown	RX2-

5.2.2 LAN2



P1



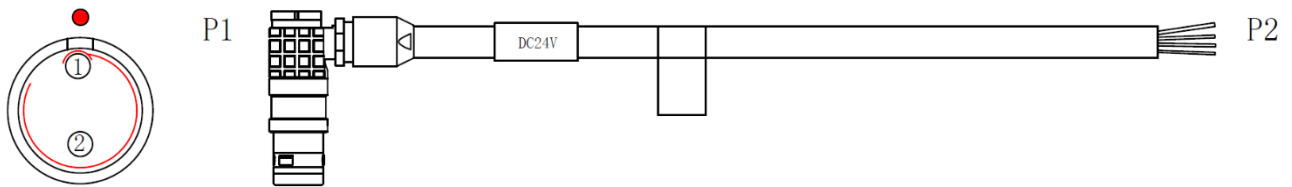
LAN2

P2

P1	Color	P2
1	Red+Light Green	+
2		
3	Black+Pink	-
4		
5	Orange	TX+
6	Yellow	TX-
7	White	RX+

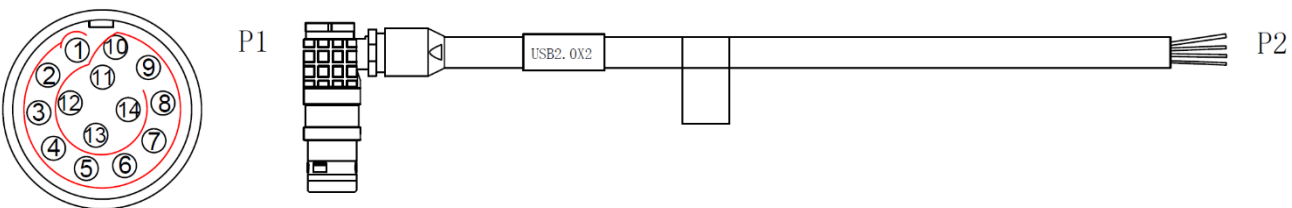
P1	Color	P2
8	Blue	TX2+
9	Purple	TX2-
10	Green	RX-
11	Grey	RX2+
12	Brown	RX2-

5.2.3 24V



P1	Color	P2
1	Brown	24V (+)
2	Blue	GND

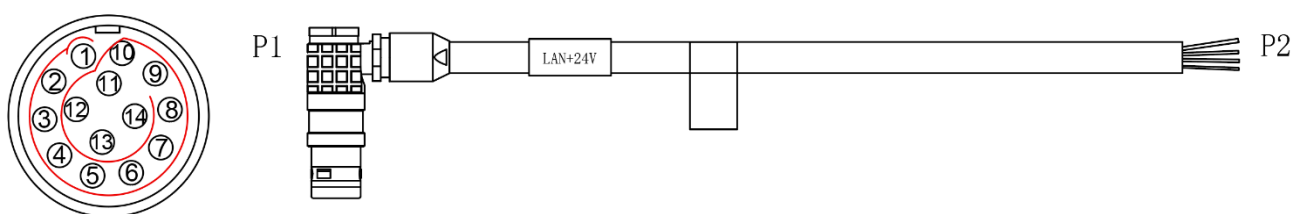
5.2.4 USB2.0×2



P1	Color	P2
1	Red+Light Green	5V-USB
2		

P1	Color	P2
3	Black+Pink	GND
4		
5	Orange	D+
6	Yellow	D-
7	White	/
8	Blue	D2-
9	Purple	D2+
10	Green	/
11	Grey	/
12	Brown	/

5.2.4 LAN5(24V) & LAN6(24V)



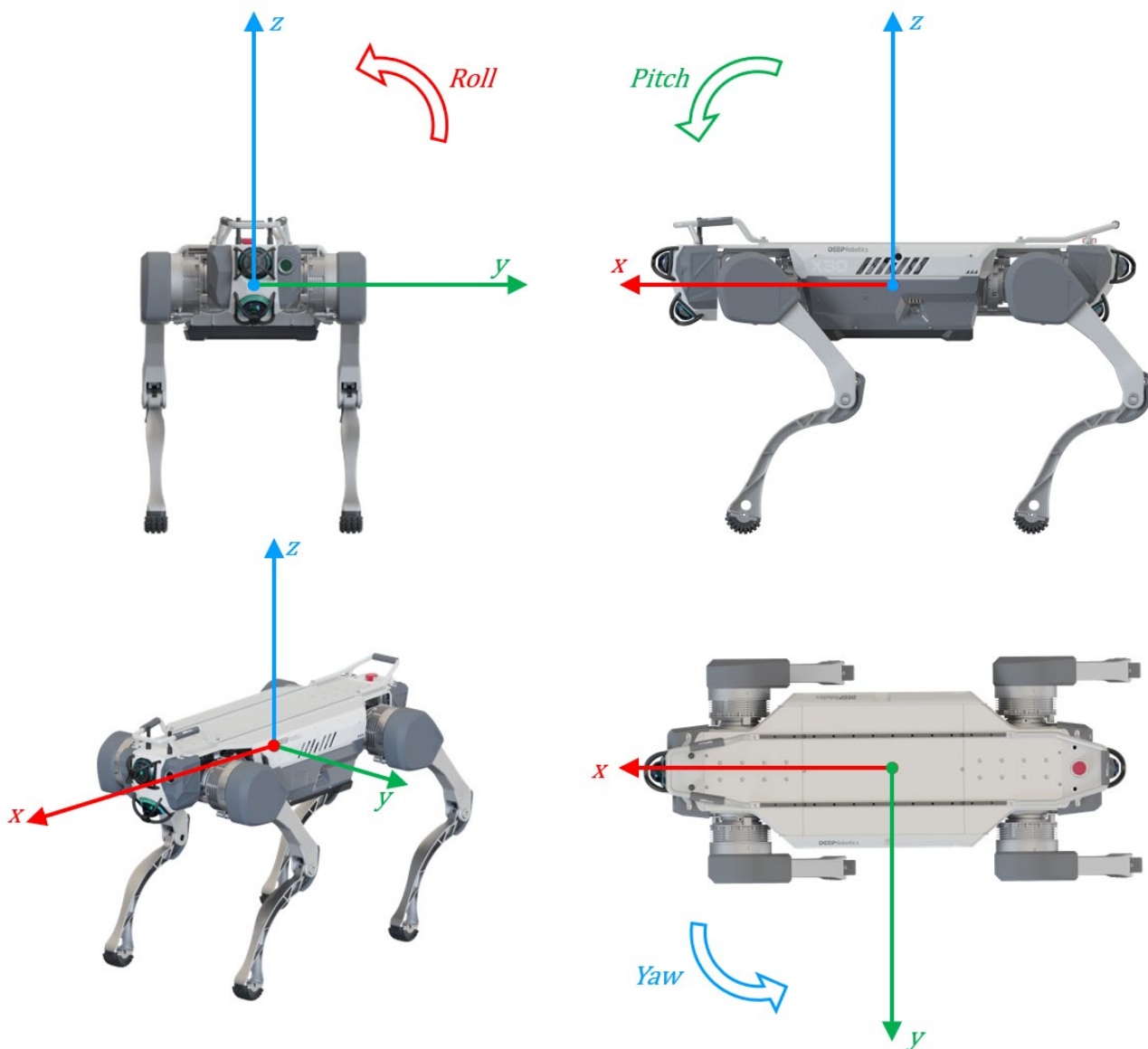
P1	Color	P2
1	Red+Light Green	24V (+)
2		
3	Black+Pink	GND
4		
5	Orange	TX+

P1	Color	P2
6	Yellow	TX-
7	White	RX+
8	Blue	TX2+
9	Purple	TX2-
10	Green	RX-
11	Grey	RX2+
12	Brown	RX2-

6 Sensor Coordinate System

6.1 Body Coordinate System

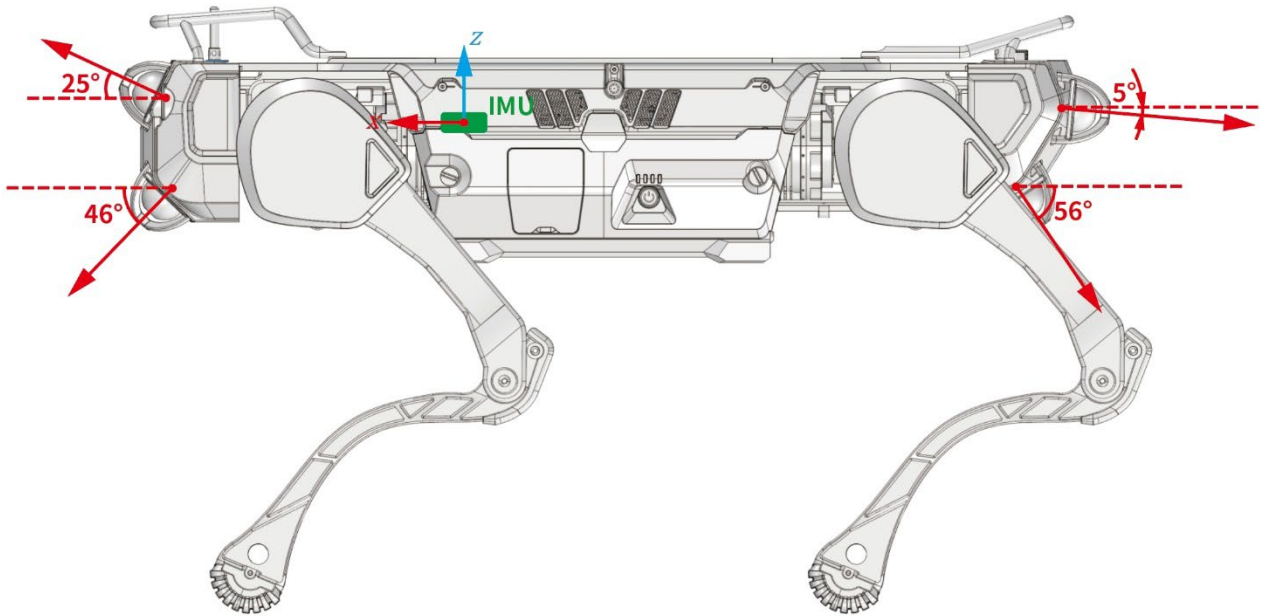
The origin of the body coordinate system is located at the geometric center of the robot (the curved arrow represents the positive rotation direction).



6.2 Sensor Coordinate Systems of Main Body

The sensors of main body include a built-in IMU and four LiDARs. The orientation of the

axes in the IMU coordinate system is consistent with that of the body coordinate system. The installation angles of the LiDARs and the orientation of the IMU coordinate system are shown below:

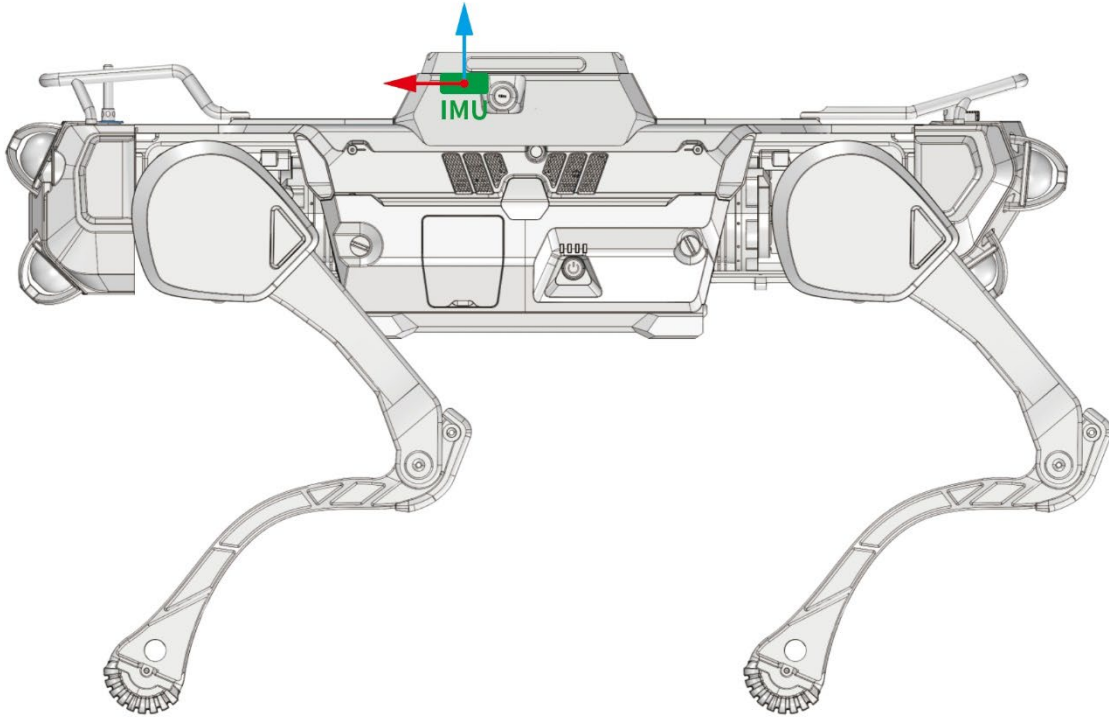


The locations of the origin of each sensor coordinate system are shown in the following table:

Sensor	Coordinates of origin of sensor coordinate system (with respect to the body coordinate system)		
	X(mm)	Y(mm)	Z(mm)
Built-in IMU	130.45	-1	14.9
Front-facing upward LiDAR	443.67	0	54.15
Front-facing downward LiDAR	432.97	0	-55.19
Rear-facing upward LiDAR	-465.76	0	34.84
Rear-facing downward LiDAR	-410.74	0	-56.01

6.3 IMU Coordinate System of Intelligent Controller

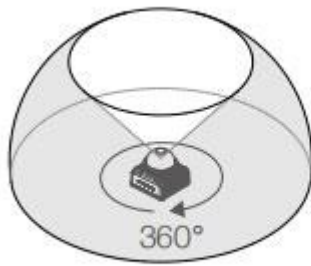
The orientation of IMU coordinate system embedded in the intelligent controller is aligned with the body coordinate system, as shown in the following diagram.



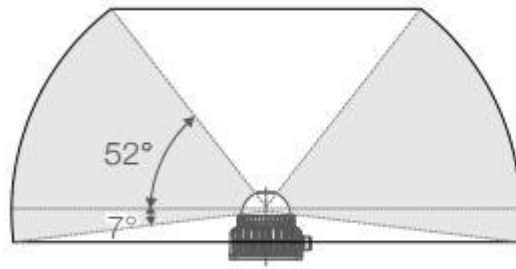
Depending on the installation position of the intelligent controller, there may be differences in the location of the origin of the embedded IMU coordinate system.

Intelligent Controller Position	Coordinates of origin of IMU coordinate system (with respect to the body coordinate system)		
	X(mm)	Y(mm)	Z(mm)
Position A	87.15	-1	117
Position B	47.15	-1	117
Position C	7.15	-1	117

6.4 LiDAR FOV



Horizontal



Vertical